Molecular Actions of PPARα in Lipid Metabolism and Inflammation

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ABSTRACT Peroxisome proliferator–activated receptor α (PPARα) is a nuclear receptor of clinical interest as a drug target in various metabolic disorders. PPARα also exhibits marked anti-inflammatory capacities. The first-generation PPARα agonists, the fibrates, have however been hampered by drug–drug interaction issues, statin drop-in, and ill-designed cardiovascular intervention trials. Notwithstanding, understanding the molecular mechanisms by which PPARα works will enable control of its activities as a drug target for metabolic diseases with an underlying inflammatory component. Given its role in reshaping the immune system, the full potential of this nuclear receptor subtype as a versatile drug target with high plasticity becomes increasingly clear, and a novel generation of agonists may pave the way for novel fields of applications. ***(Endocrine Reviews 39: 760 – 802, 2018)***

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The peroxisome proliferator–activated receptors (PPARs) are ligand-inducible transcription factors that belong to the nuclear receptor (NR) superfamily. Three PPAR subtypes have been identified: PPARα (NR1C1), PPARβ/δ (NR1C2), and PPARγ (NR1C3) (1–5).

The first member of the PPAR family, PPARα, was identified in 1990 and named by its ability to become activated by chemicals known to induce peroxisome proliferation in rodents (6). Peroxisomes are subcellular organelles found in most plant and animal cells and harbor a complex set of enzymes acting on various metabolic functions, including β-oxidation of fatty acids (FAs), bile acid, and cholesterol metabolism (7).

Peroxisome proliferators cover a various set of synthetic and structurally diverse industrial and pharmaceutical chemicals, including herbicides, industrial solvents, and hypolipidemic drugs, that cause significant increases in the size and number of peroxisomes in liver, liver hypertrophy, liver hyperplasia, and hepatocarcinogenesis in rodents (8). Following the initial discovery of mouse PPARα, the two other PPAR subtypes, PPARβ/δ and PPARγ, were identified by homology screens (2, 9). Contradicting their name, the latter two subtypes do not cause peroxisome proliferation in mice (8). The PPAR subtypes are encoded by distinct genes located on different chromosomes [reviewed in Ref. (10)] and share a high degree of structural and sequence homology, but each subtype displays a divergent pattern of tissue-specific expression and ligand-binding specificity (9, 11).

PPARs were originally described as orphan NRs, but shortly thereafter a plethora of potential endogenous ligands were described (12), acknowledging PPARs to be activated by a wide range of endogenous ligands including FAs and FA metabolites, such as hydroxylated eicosanoids, prostaglandins, and leukotrienes, as well as by many synthetic compounds, including fibrates and antidiabetic thiazolidinedione drugs (10, 13).

Most of the physiological functions of PPARs can be explained by their activity as transcription factors

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ESSENTIAL POINTS

- Peroxisome proliferator–activated receptor (PPAR)α is a transcription factor whose levels and activity are controlled at many different steps (*e.g.*, gene and protein expression and protein posttranslational modifications) and by many different factors (*e.g.*, nutritional/hormonal cues, aging, cytokines, growth factors)
- Natural ligands exist and many different synthetic ligands have been synthesized, some only for basic research but some currently in clinical phase III trials for human disease
- PPARα drives PPAR response element–dependent and –independent gene transcription through various molecular mechanisms, in support of the versatility of PPARα and expanding its target gene repertoire
- PPARα target genes are not only involved in glucose and lipid metabolism but also in inflammation-modulating pathways
- A most recently developed high-affinity PPARα agonist is currently tested as a potential treatment of the inflammatory state of nonalcoholic fatty liver disease in humans, that is, nonalcoholic steatohepatitis
- Although the vast majority of PPARα's activities yield an anti-inflammatory profile, inflammation-priming effects have also been documented; the circumstances in which the latter events occur and the physiological relevance hereof remain to be further clarified
- Future therapeutics relying on PPARs in a context of inflammation will most probably have to take "the clock" into consideration

that modulate the expression of specific target genes (14–17). PPARs can bind to sequence-specific target elements in the promoter region of target genes following heterodimerization with the retinoid receptor to drive several aspects of normal cellular physiology as well as pathology (15, 18). This includes lipid metabolism, glucose homeostasis, cell differentiation, proliferation, obesity, cancer, and inflammation (19, 20). In addition to positively regulating gene expression, activated PPARs can inhibit gene expression by negatively interfering with the activity of proinflammatory transcription factors, including signal transducer and activator of transcription (STAT), activator protein-1 (AP-1), and nuclear factor-κB (NF-κB) (21–23). Such transrepression mechanisms are likely to participate in the anti-inflammatory actions of PPARs (24).

In this review, we focus mainly on the molecular biology of the PPARα subtype and its role in inflammation control and inflammation-related disorders. The focus is on the liver, adipose tissue, and cardiovascular and immune systems as primary tissues/organ systems of interest.

PPARα Expression Profile, Gene, and Protein Structure

PPARα controls the expression of numerous genes involved in a plethora of lipid metabolic pathways, including microsomal, peroxisomal, and mitochondrial FA oxidation (FAO), FA binding and activation, FA elongation and desaturation, synthesis and breakdown of triglycerides (TGs) and lipid droplets, lipoprotein metabolism, gluconeogenesis, bile acid metabolism, and various other metabolic pathways and genes (25). Consistent herewith, PPARα is widely expressed in tissues with high FAO rates, such as heart, liver and skeletal muscle, and functions as a major regulator of FA homeostasis (26–28). PPARα expression is also significant in brown adipose, adrenal, and kidney tissue, as well as in several inflammatory/immune cell types such as monocytes/macrophages (26, 27, 29).

It is generally agreed that active PPARα is predominantly localized in the nucleus (30). However, PPARα was reported to be localized in the cytoplasm of rat chondrocyte primary cultures (31). Similarly, in differentiated human macrophages, PPARα was also much more localized in the cytoplasm (32). Hence, as with other NRs, PPARα displays a dynamic shuttling between the cytosol and the nucleus. This shuttling is influenced by Ca^{2+} signaling and ligand binding, with the latter enhancing nuclear import of PPARα (33). In this way, internal and/or external cell signals can be translated into changes in subcellular localization and turnover/stability of PPARα and, as a consequence, affect its regulation of cellular functions (33). The observation of a nucleocytoplasmic shuttling is not only restricted to PPARα, but also applies to PPARγ. MAPK kinase–dependent export of PPARγ from the nucleus reduces its transcriptional activity and therefore represents a mechanism by which nuclear activities of PPARγ and possibly also other NRs are downregulated (34). Further investigations are required to fully elucidate the underlying molecular mechanism of nucleocytoplasmic shuttling as well as the subcellular functions of this receptor. Moreover, a better understanding of where and how PPARs are activated, turned over, and regulated is of the utmost importance.

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The human and mouse PPARα genes, located on chromosome 22 and chromosome 15, respectively, encode 468 amino acid polypeptides with 91% homology (35). As in the mouse, the human PPARα gene–encoded mRNA is derived from eight exons with a 5′ untranslated region (UTR) encoded by exons 1, 2, and part of exon 3. The remainder of exon 3 and exons 4 to 8 contribute to the coding region of PPARα (35, 36) [Fig. 1(a)]. The last 232 bp of exon 8 contribute to the 3′ UTR region (37). In correspondence with other NRs and PPARs, the encoded PPARα protein displays four distinguishable functional domains whose cooperation modulate receptor activity (38). These domains are an N-terminal region (A/B domain), a central DNA-binding domain (DBD, C domain), a flexible hinge region (D domain), and a C-terminal ligand-binding domain (LBD, E domain) (39, 40) [Fig. 1(b); Table 1 (33, 35, 38, 40–57)]. The N-terminal A/B domain, which is the least conserved domain among PPARs, harbors a ligand-independent activation function-1 (AF-1) region. This region, enriched with acidic amino acids, has a low level of basal transcriptional activity and can function independently of ligand binding (tested via GAL4-fusion proteins in one-hybrid assays) (35, 40). Although the A/B domain suffers from a poor structural organization, secondary structure formation in this domain is suggested as an important step toward AF-1–mediated transactivation (38, 40). The importance of the A/B domain is emphasized by the fact that its deletion results in a gene-dependent alteration in PPARα transcriptional activity. For example, A/B domain deletion interrupts PPARα-mediated transactivation of the acyl–coenzyme A (CoA) oxidase (ACOX)

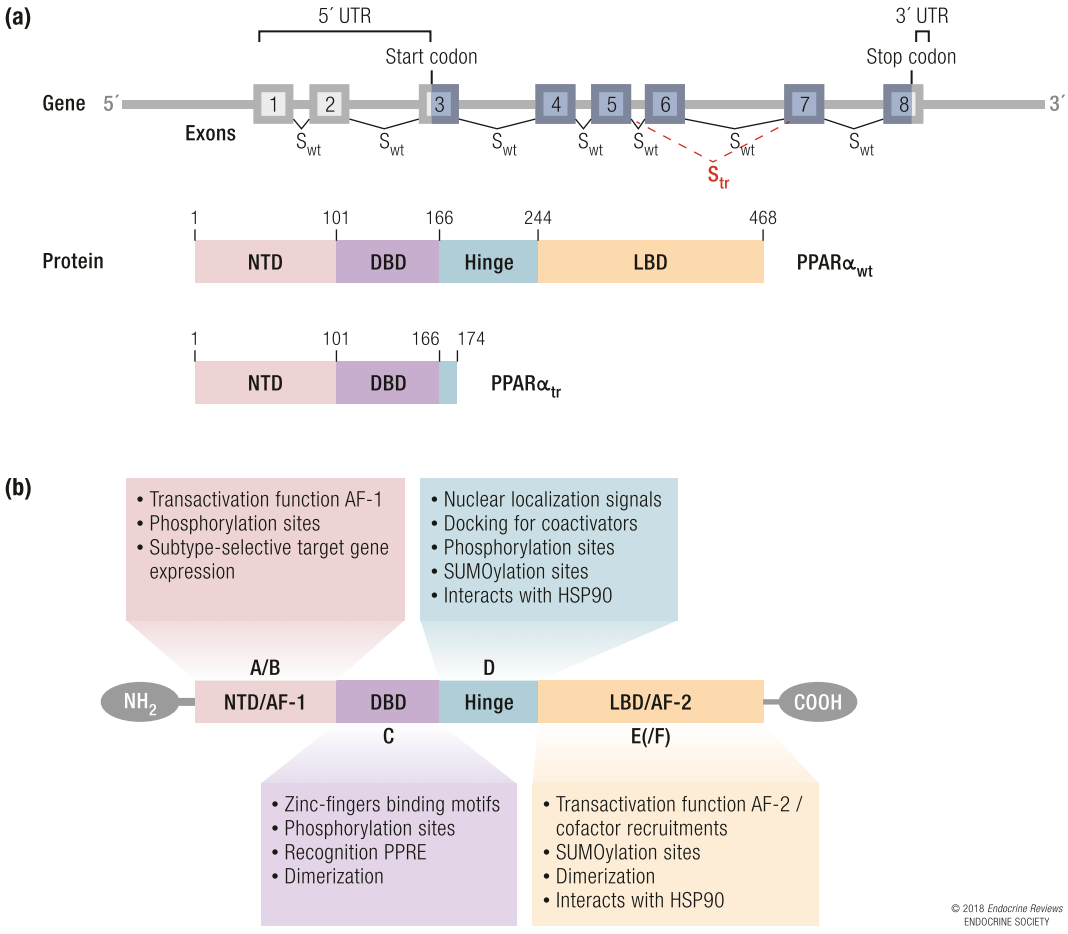


Figure 1. Schematic presentation of the human PPARα gene and protein domain structure–function relationship. (a) The human PPARα gene contains at least eight exons. Exons are shown in boxes with translated regions shaded. The 5′ UTR is encoded by exons 1, 2, and the 5′ end of exon 3. The coding DNA sequence spans the 3′ region of exon 3, exons 4 to 7, and the 5′ extremity of exon 8. The 3′ UTR consists of the last 232 bp of exon 8. The splicing events S_{wt} (wild-type) and S_{tr} (truncated) generating hPPARα$_{wt}$ and hPPARα$_{tr}$ transcripts, respectively, are depicted. The variant transcript lacks exon 6, resulting in a frame shift generating a premature TGA termination codon. The corresponding protein products of the two splice isoforms are shown beneath the gene. (b) PPARα contains four distinct domains. The amino-terminal A/B domain contains the ligand-independent AF-1, the C domain consists of a highly conserved DBD, the D domain consists of a variable hinge region, and the C-terminal E domain includes an LBD and a ligand-dependent AF-2. Per domain, main functions are listed.

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Table 1. PPARα Functional Domains and Their Relative Functions

Domain	Functions	Refs.
N-terminal (A/B domain)	Transactivation function AF-1	(35, 40)
	Phosphorylation sites	(41–43)
	Subtype-selective target gene expression	(38, 44)
DBD (C domain)	Zinc finger binding motifs	(45)
	Phosphorylation sites	(46)
	Recognition PPRE	(47, 48)
	Dimerization	(49)
Hinge (D domain)	Nuclear localization signals	(33, 50)
	Docking for coactivators	(51)
	Phosphorylation sites	(46, 52)
	SUMOylation sites	(53)
	Interacts with Hsp90	(54)
C-terminal LDB (E/F domain)	Transactivation function AF-2/cofactor recruitments	(55)
	SUMOylation sites	(56)
	Dimerization	(51, 57)
	Interacts with Hsp90	(54)

PPARα displays a classic NR canonical structure. Per domain, main PPARα functions are listed (details in the text). Abbreviation: PPRE, PPAR response element.

promoter; however, transactivation of the cytochrome P450 4A6 promoter is unaffected (40, 45). Additionally, AF-1 has also been proposed to be a key determinant of subtype-selective target gene expression among the PPAR subtypes. For example, addition of the PPARα A/B domain to PPARγΔAB (A/B domain truncated) enhances its ability to activate PPARα-specific target genes (38). Oppositely, addition of the PPARγ A/B domain to the nonadipogenic PPARβ/δΔAB was sufficient to provide adipogenic potential to the resulting PPARβ/δ chimeric protein (44). Posttranslational modifications (PTMs) further fine-tune PPARα functionality. PPARα AF-1 domain phosphorylation leads to changes in transcriptional activity and even ligand binding of the receptor (43). For example, MAPK-mediated phosphorylation of serine 12 and 21 in the human PPARα A/B domain increases the transcriptional activity in response to mitogenic stimuli, such as high concentration of insulin (58). Alternatively, glycogen synthase kinase 3 (GSK3)–mediated phosphorylation of serine 73 increases the ubiquitination and degradation of PPARα (42). Concerning its relationship/association with other domains in the protein, PPARα AF-1 synergistically activates transcription together with AF-2, as found for other NRs (40).

The DBD, which is highly conserved and rich in cysteine and basic amino acids, consists of two zinc-finger binding motifs (45). This domain is responsible for physical interaction with DNA. Similar to other PPAR subtypes, PPARα binds DNA as an obligate heterodimer with retinoid X receptors (RXRs) (NR2B) (49). The receptor can be activated not only by phosphorylation of the A/B domain or via ligand binding to PPARs themselves, but also via associating with RXR, which for example binds 9-*cis* retinoic acid. The latter activation mechanism is referred to as the permissive effect. Ligand binding of either receptor thus activates the PPAR–RXR complex; however, simultaneous binding of both ligands can lead to a more potent or even synergistic activation (59–65). These different activation mechanisms allow fine-tuning of PPAR activity.

The PPARα–RXR heterodimer recognizes and binds to PPAR response elements (PPREs), which are localized in gene regulatory regions and organized as a direct repeat type 1 (DR-1). DR-1 consists of two copies of the hexameric nucleotide recognition motif 5′-AGGTCA-3′ separated by a single nucleotide (47, 48). An additional consensus AACT motif is positioned 5′ to the DR-1 (66, 67). Binding to the PPRE occurs in such a way that PPARα binds to the 5′ extended half-site of the response element, whereas RXR occupies the 3′ half-site (5, 66, 68). The PPARα–RXR heterodimers can form independently of the PPARα ligand. These unliganded heterodimers recruit the corepressor protein complex and inhibit target gene transcription (69). Upon ligand binding, the corepressor complex is

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released from the PPARα–RXR heterodimer and the coactivator complexes will be recruited to the promoter region of target genes, thereby initiating transcription (70) [Fig. 2(a)].

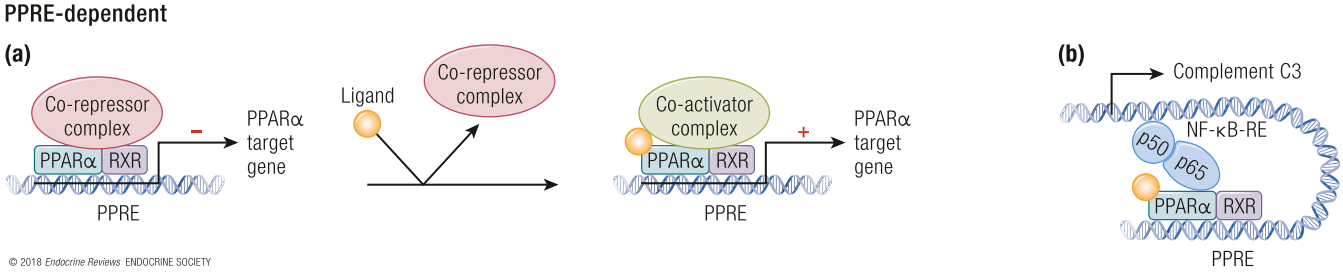
The first natural PPRE was found in the promoter region of the ACOX gene (36, 71). Additionally, recent genome-wide profiling of PPARα binding sites revealed ~46% of PPARα–RXR binding sites within intronic regions (72). Using PPARγ as the study case, chromatin accessibility in general was identified as a major determinant for transcriptional activity, as PPAR–RXR heterodimers were found to primarily bind in open chromatin regions (73–75).

Whereas PPARα binds DNA only as a heterodimer and not as a monomer, deletion of its N-terminal A/B domain generates a truncated protein that binds DNA nonspecifically as a monomer in *in vitro* assays (45). Although the physiological significance of such binding is not completely understood, it may serve as evidence of an interdomain communication. The DBD of PPARγ was shown to form an interface with its own LBD as well as with the LBD of its heterodimeric partner, thereby influencing ligand binding (76). More crystallization efforts are needed to challenge or validate this interface concept in the case of PPARα. The DBD of PPARα (and PPARγ) is also subject to phosphorylation, which serves to further modulate the transcriptional activity of the receptor (46).

Similar to PPARs, many other NR members also need RXRs as their obligatory heterodimerization partner, raising therefore the possibility of a competition between particular NRs for their common binding partner (77). This competition may have important biological outcomes. For instance, liver X receptor (LXR)α activation by either overexpression of LXRα or ligand activation suppresses PPARα signaling through reduction of active PPARα–RXR heterodimers in the liver and thus diminished PPRE occupancy. RXRα supplementation restored the LXR-mediated inhibition of PPARα activity (78). Conversely, excess PPARα in the presence of its ligands suppresses the sterol regulatory element–binding protein (SREBP)-1c promoter that contains two LXR response elements (79), indicating that the mechanism of competition for RXRα between PPARα and LXR may be reciprocal (78). This relationship between PPARα, LXR, and RXR could be crucial for a mutual regulation of both LXR and PPARα activities, and thus for the nutritional regulation of their downstream target genes. However, the physiological validation of these mechanisms still remains to be demonstrated. Another example of competition for RXRα between RXR-partnering NRs was described for the thyroid hormone receptor with LXR (80).

Of note, silent information regulator 1 (SIRT1) was also shown to bind PPARα through a region homologous to the PPARα binding domain in RXRα. Interestingly, the balance between SIRT1 and RXR appears to determine whether PPARα is a positive or negative regulator of genes involved in FA metabolism (81). Additionally, the PPARα–SIRT1 complex would directly bind a single hexameric motif embedded in estrogen-related receptor (ERR) elements, thus competitively downregulating ERR target genes, at least in the heart. These findings suggest a PPARα-dependent transcriptional repression mechanism in the control of ERR-driven mitochondrial respiration and cardiac contraction (82, 83). Additionally, exposure to a high-fat diet (HFD) also induces the formation of a PPARα–SIRT1 complex, impairing an association of PPARα with the proinflammatory transcription factor p65. Collectively, this shift is expected to lead to the repression of PPARα-target genes involved in FAO and enhancement of proinflammatory pathways in the heart (84). Albeit intriguing, more experimental evidence is necessary to prove that PPARα can bind DNA as an RXR-independent monomer and whether this mechanism is also operative in other tissues such as the liver.

Immediately adjacent to the DBD is the highly flexible hinge domain (D), connecting the DBD and



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Figure 2. PPRE-dependent transcription regulation. (a) PPRE-dependent activation. In the absence of specific ligands, PPARα/RXR heterodimers bind to the PPRE located in the promoter of target genes, or elsewhere in the genome, and recruit corepressor complexes, thereby maintaining PPARα target genes in a silent state (ligand-independent repression). Upon ligand activation, the nuclear corepressors are released and the coactivator complexes are then recruited and bound to PPARα/RXRα heterodimers, hence inducing expression of target genes involved in several biological processes, including glucose and lipid metabolism. (b) PPRE-dependent inhibition of NF-κB transcriptional activity. Upon ligand activation, PPRE-bound PPARα directly interferes with the activity of p65, antagonizing thereby its binding to an NF-κB RE in the complement promoter.

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the LBD (35). This region binds corepressor proteins, such as the NR corepressor (NCoR), to the receptor in its inactive, unliganded state (51) and serves as a target for PTMs, including phosphorylation on serines 179 and 230 catalyzed by protein kinase C (PKC) and the SUMOylation at Lys185 that promote NCoR recruitment (52, 53). Additionally, mutation of a potential phosphorylation site within the PPARα hinge was shown to prevent dimerization capacities (46). Another study on truncated PPARα lacking part of the hinge region and LBD suggested that the hinge region might contain a nuclear localization signal (50). Consistently, a series of truncation mutants revealed that PPAR nuclear transport is mediated by at least two nuclear localization signals, that is, one in the DBD-hinge region and one in the AF-1 domain (33). It was suggested that the hinge region also interferes with PPARα binding to DNA and hence takes up an important suppressor role (85). More specifically, the ribosomal protein L11 was shown to interact with the D-domain of PPARα and to inhibit PPRE-driven reporter activity (85). Furthermore, heat shock protein 90 (Hsp90) was reported to associate with both the hinge and LBD of PPARα. Using several approaches to disrupt Hsp90 complexes within the cell, Hsp90 was confirmed as a repressor of PPARα (and PPARβ/δ) activity (54).

Next, the PPARα LBD harbors a ligand-dependent transactivation function (AF-2) and has a structure consisting of an α-helical sandwich and a four-stranded β-sheet, forming a Y-shaped cavity of 1400 Å called the ligand-binding pocket (LBP) (55). The volume of the PPARα LBP is quite comparable to other PPAR subtypes, but it is substantially larger as compared with most other NRs. Among the PPARs, the PPARα LBP is more lipophilic, which potentially explains the greater affinity of PPARα to bind the more saturated FAs (55). Following ligand binding, the AF-2 domain undergoes conformational changes, which allows for the interaction with various coactivators carrying LXXLL motifs (L, leucine; X, any amino acid) (55), such as cAMP response element binding protein (CBP)/p300 and steroid receptor coactivator-1 (SRC-1/NCoA-1) (55). Additionally, the LBD is crucial for dimerization with RXR; this was confirmed in studies involving deletion of PPARα helix 10 to 12 as well as an L433R mutation in PPARα, which caused impaired heterodimerization with RXR (51, 57). PPARα furthermore possesses two nuclear export signals in the DBD and LBD regions, which are recognized by the export receptors calreticulin and chromosomal maintenance 1, respectively. A recent study confirmed the presence of a functional nuclear export signal in the LBD, that is, at amino acids 300 to 308 in human PPARα (86, 87).

As mentioned earlier, human and murine PPARα show 85% identity at the nucleotide level and 91% identity at the amino acid level; however, evidence exists for a genetic heterogeneity in the functional coding sequence of human PPARα that translates into functional differences in receptor activity. For example, PPARα L162V, a variant of the human PPARα gene resulting in the introduction of an amino acid substitution within the PPARα DBD, is found at an allelic frequency of ~0.025 to 0.073 in ethnically diverse populations (88–90), whereas it is more frequent (0.745) in North Indians (91). Functional studies on the L162V gene variant demonstrate enhanced ligand-induced activity when compared with the wild-type (88, 91). Importantly, in various studies the L162V polymorphism is correlated with dyslipidemia, coronary ischemic events (37, 88–90, 92–94), and adiposity (95, 96). However, other studies have not replicated these correlations (97, 98). These inconsistencies could be due to differing environments, genetic background, and disease status between the studied populations. In this context, the effect of the L162V polymorphism could be modulated via gene–drug and gene–nutrient interactions (94, 99, 100). For example, the association of the L162V genotype with a particular lipoprotein profile may be modified by dietary fats (94, 101).

V227A, a natural variant in which the PPARα hinge region is affected, was identified in a Japanese population with a frequency of 5.0% (102–104) and in a Chinese population with a frequency of ~4% (105). This variant attenuates the PPARα-mediated transcription of cytochrome P450 4A6 and the mitochondrial 3-hydroxy-3-methylglutaryl–CoA (mHMG-CoA) synthase genes in the presence of fibrates. This transcriptional repression was ascribed, at least partially, to increased corepressor binding to PPARα and defective release of NCoR from chromatin (106). Interestingly, the V227A variant was associated with perturbations in plasma lipid levels and modulated the association between dietary polyunsaturated FA intake and high-density lipoprotein (HDL) cholesterol (HDL-C) concentrations (105). Of note, the PPARα V227A polymorphism may be involved in the pathogenesis of nonalcoholic fatty liver disease (NAFLD) and may play a protective role in obesity (107). Although epidemiological and interventional studies demonstrate a link between PPARα polymorphisms and metabolic parameters, more studies are required to understand whether and how the response to fibrate drugs and other PPARα-regulated pathways are affected by the polymorphisms.

In addition to polymorphic variants, a truncated splice variant of PPARα, generated by alternative splicing resulting in the skipping of exon 6, has been detected in human PPARα [Fig. 1(a)] (50, 108). This variant lacks the hinge region and the entire LBD and is unable to bind to PPREs. As a consequence, it exerts a dominant-negative transcriptional repressive activity on full-length PPARα through titration of the coactivator CBP by competing for coactivators. It was speculated that differences in the expression level of

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truncated human PPARα (hPPARα) might be implicated in the heterogeneity in response to fibrates among different patients (50). Of interest, genome-wide correlation analysis with subsequent pathway enrichment analysis indicated a selective role for truncated PPARα as an antiproliferative and anti-inflammatory factor (109). Further experimental manipulation of human and mouse hepatocytes by specific knockdown and overexpression analysis substantiated this hypothesis (109). *In vivo* evidence for a dominant-negative transcriptional repressive activity is lacking so far. Therefore, whether this variant has any specific physiological significance with regard to metabolic processes, inflammation, as well as its relevance for hepatocarcinogenesis still needs to be revealed and requires further investigation.

Modulating Factors of PPARα Activity

PPARα gene and protein expression

The expression of the PPARα gene is regulated by various physiological conditions such as stress, glucocorticoids (GCs), growth hormones, insulin, and leptin. As an important physiological response, PPARα expression is induced during fasting, resulting in an enhanced expression of PPARα target genes in liver to operate FAO and to form substrates that can be metabolized by other tissues. Oppositely, PPARα also controls, directly or indirectly, lipogenic pathways (which is further discussed in detail in "PPARα Target Genes and Their Function"). Liver lipogenesis particularly allows FA synthesis in a fed state when dietary carbohydrates are abundant (110, 111). Adipocyte-derived factors may further influence the expression level and subsequent PPARα activity. Consistent with this, administering leptin, considered proinflammatory, correlated with an increased PPARα hepatic expression in *ob*/*ob* mice (112). The mechanisms underlying the induction of PPARα transcription by leptin may be connected to changes in FA flux mediated by leptin signaling (112). Leptin-stimulated PPARα gene expression was also observed in muscle cells, through the activation of and changes in the subcellular localization of AMP-activated protein kinase (AMPK)-α2 (113). Additionally, leptin was found to enhance PPARα gene transcription in pancreatic β-cells (114) but not in adipose tissue (115). The basis for this discrepancy is not known, but findings generated using isolated model systems will definitely require further confirmation *in vivo*.

Alternatively, the anti-inflammatory adipokine adiponectin also increased PPARα-responsive promoter activity in $H4IIEC_3$ cells, an effect that required coexpression of PPARγ coactivator (PGC)-1α, suggesting that adiponectin-induced PPARα activity may be mediated through upregulation of PGC-1α expression (116). In contrast, GH has been shown to rather decrease PPARα mRNA in cultured rat hepatocytes (117, 118). A similar effect was shown in hypophysectomized rats (119). Additionally, GH also decreases PPARα transcriptional activity (120) as well as the expression level of PPARα (121). Similarly, treatment of primary hepatocytes for 3 days with another anabolic signal, insulin, also led to reduced PPARα levels (122), whereas short-term exposure to this hormone increased PPARα transactivation via phosphorylation of the ligand-independent AF-1 domain (58).

Next to tissue-specific differences in PPARα levels, the expression of PPARα in a given tissue further oscillates under the control of the circadian rhythm. For example, PPARα expression peaks during daytime in mice, which correlates with the timing of fasting in these nocturnal rodents (123). These diurnal patterns thus correspond well with known functions of PPARα in physiology but also pathophysiology (124). PPARα is under the transcriptional control of the GC-activated GC receptor (GR) in rat hepatocytes (125). As such, PPARα expression follows a diurnal rhythm, which parallels that of circulating corticosterone, and the levels of PPARα are also upregulated during times of stress (126). Moreover, the circadian expression of PPARα was reported to be directly regulated by the core-clock transcription factors brain and muscle Arnt-like (bmal) protein and circadian locomotor output cycles kaput (CLOCK) both *in vivo* and *in vitro* (127). In another study, the proline- and acidic amino acid–rich basic leucine zipper (PARbZip) proteins were also shown to modulate the rhythmic expression and activity of PPARα. The mechanism was proposed to be indirect and involved a cyclic generation of PPARα ligands that initiate a feed-forward loop, in which PPARα enhances transcription from its own gene (128). Additionally, enhanced PPARα transcript levels were observed in both B and T cells upon GCs treatment (129), indicative of a function of PPARα in the immune system (see "PPARα Agonists in a Clinical Setting").

Additionally, PPARα expression is also reported to be related to aging. In this line of reasoning, examination of mRNA expression in mouse spleens revealed an age-associated decline in the levels of mRNA encoding PPARα and its concomitant target genes ACOX and catalase, which was accompanied by a constitutive proinflammatory cytokine production (130). Similarly, a recent study has shown that aging inhibits PPARα activation as assessed by diminished mRNA levels of its target genes ACOX and carnitine palmitoyl transferase 1 (CPT1) in aged rat livers (131).

Metabolite levels also affect PPARα expression; intriguingly, glucose and metabolites (glucose-1-phosphate and glucose-6-phosphate) were shown to bind PPARα with high affinity, resulting in a significantly altered PPARα secondary structure (132). In this study, *in*

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vitro addition of glucose alone reduced the interaction of PPARα with its coactivator SRC-1. Conversely, in the presence of the activating ligands arachidonic acid and clofibric acid, glucose rather increased PPARα interaction with SRC-1, DNA binding, and activation of the β-oxidation pathway (132). Additionally, high glucose levels regulate PPARα expression in human macrophages. Incubation of murine J774 macrophages with high glucose concentrations increased the expression of PPARα at the mRNA and protein levels and enhanced its binding to the PPRE of the lipoprotein lipase (LPL) promoter (133). In contrast, glucose decreases PPARα expression in pancreatic β-cells, leading to a reduced FAO and increased phospholipids and TG synthesis, a mechanism thought to underlie β-cell glucolipotoxicity (134).

NF-κB–driven cytokines such as TNF-α, IL-1β, and IL-6 also affect the expression of PPARα. In TNF-α–treated male Sprague Dawley rats, PPARα mRNA and corresponding PPARα protein levels were significantly reduced, likely contributing to the decrease of peroxisomal β-oxidation (135). IL-1β and IL-6 both repress the induction of PPARα expression exerted by the combined action of clofibric acid and the GC dexamethasone in fetal rat hepatocytes (136).

The presence or absence of the activity of other transcription factors also impacts PPARα levels and activity. In hepatocytes, Krüppel-like factor 6 deficiency reduces PPARα protein expression and downstream targets, resulting in attenuation of lipid and glucose abnormalities associated with an HFD. Krüppel-like factor 6 mediated induction of PPARα protein expression through miRNA-10b repression (137). PPARα is also regulated at the transcriptional level by other NRs such as hepatocyte nuclear factor 4 (HNF4), chicken ovalbumin upstream promoter–transcription factor II, and others. Whereas HNF4 positively regulates hPPARα expression via a DR-1 element present in the hPPARα promoter, chicken ovalbumin upstream promoter–transcription factor II also bound this site but antagonized this effect (138). The regulation of PPARα by HNF4, consistent with the tissue-expression pattern of PPARα, was also demonstrated *in vivo* using HNF4-deficient mice that display significantly lower PPARα mRNA levels compared with the wild-type mice (139). As an example of another metabolite, bile acids stimulate hPPARα expression in a species-specific manner through a mechanism involving the binding of farnesoid X receptor on its promoter (140). Moreover, PPARα expression in the mouse small intestine was shown to be under the control of LXR and the pregnane X receptor (141).

Cross-talk with other NR signaling pathways may also modulate PPAR activity, as illustrated by the direct control of the PPARβ promoter by the vitamin D receptor (142). In this respect, and as mentioned earlier, NRs that function as heterodimers with RXR can potentially compete with PPARα signaling in tissues in which the amount of RXR is limiting (77).

Furthermore, PPARα turnover is controlled by ligand specificity and availability (Fig. 3). In this context, Blanquart *et al.* (41) demonstrated that synthetic PPARα ligands such as Wy14,643, GW7647, or fibrates prolonged half-life and increased stability of the receptor, preventing thereby its ubiquitination and its subsequent degradation via the proteasome. Besides, PPARα agonists, such as fenofibrate, markedly upregulated hPPARα mRNA levels in primary cultures of human hepatocytes as well as hPPARα promoter activity, suggesting that PPARα positively autoregulates its own expression (138).

Of note, hPPARα mRNA levels in the liver vary among individuals (50, 109), suggesting that PPARα is strongly regulated at the gene level in human liver by genetic and/or environmental factors (50).

Lastly, the function of PPARα as a transcription factor is controlled by its PMTs and cofactor recruitment (Fig. 3), which is further discussed in detail in the next sections.

Taken together, regulation of the PPARα expression levels efficiently modulates PPARα activator responses and thus may affect downstream pathways involved in metabolism and inflammatory processes.

"Finding a way to regulate the receptor activity through modulating endogenous ligands might generate new therapies."

PPARα cofactors

As with other NRs, PPARα transactivation activity is regulated by the presence of coregulators, which may function as corepressors or coactivators depending on the context. In the absence of a specific ligand, PPARα binds corepressors, such as NCoR and silencing mediator for retinoid and thyroid (SMRT) hormone receptors (26). The receptor interacting protein 140 (RIP140) is a PPARα corepressor, which is known to repress the activity of NRs by competing with coactivators and by recruiting downstream effectors such as histone deacetylases (143). Interestingly, mice devoid of the corepressor protein RIP140 show resistance to HFD-induced obesity and hepatic steatosis in association with upregulation of genes involved in energy dissipation. Such phenotype may be suggestive of a role for this corepressor in PPARα signaling (144). Whether PPARα is directly involved remains to be shown. Alternatively, known coactivators that have been found in a large complex with PPARα–RXR heterodimers include PGC-1α, PGC-1β, CBP/p300, SRC-1, PPAR-interacting protein, PPAR-binding protein (PBP/TRAP220/DRIP205/MED1) as well as coactivator-associated proteins PIMT (NCoA6IP) and coactivator-associated arginine methyltransferase-1. Some of the coactivators, including the SRC-1 family and CBP/p300, contain intrinsic histone acetyltransferase activities required to facilitate chromatin remodeling [reviewed in Ref. (145)]. The PPARα-interacting cofactor complex isolated from rat liver nuclear extracts reveals the presence of many of these

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coregulators, likely forming one megacomplex (146). Most cofactors are not receptor exclusive; however, emerging gene knockout mouse models show that some of the coactivators are essential for PPARα signaling. For example, conditional deletion of PBP/MED1 gene in liver results in the abolition of PPARα ligand–induced pleiotropic effects, indicating that PBP/MED1 is essential for the function of PPARα (147).

PPARα PTMs

As for other NRs, PPARα activity is finely tuned by various PTMs. including phosphorylation, ubiquitination, and SUMOylation (41, 53, 148).

PPARα is a phosphoprotein of which the transcriptional activity is affected by cross-talk with several kinases, including MAPK (ERK1/2), c-Jun N-terminal kinases/p38, AMPK, protein kinase A (PKA), PKC, and GSK3 [Table 2 (42, 52, 58, 113, 148–153)].

The effects of phosphorylation on PPARα activity depend on a number of features such as the nature of the modified residue, as well as the stimulus and the kinase involved. PPARα phosphorylations may also have different functional outcomes depending on the context, with effects on receptor stability, susceptibility to proteosomal degradation, enhanced recruitment of cofactors, and transactivation. PPARα was first shown to be phosphorylated in primary rat adipocytes in culture in response to insulin, resulting in the stimulation of basal as well as ligand-dependent transcriptional activity (148). Along the same line of interest, insulin treatment of human HepG2 hepatoma cells increases PPARα phosphorylation, enhancing thereby its transcriptional activity. This insulin-mediated phosphorylation of PPARα via the ERK1/2 MAPK pathway involves amino acids Ser12 and Ser21 in the AF-1 domain, resulting in transcriptional activation probably by relieving inhibition by corepressors (42, 58). In line with this, in rat neonatal cardiac myocytes, stress-activated p38 MAPK phosphorylates Ser6, Ser12, and Ser21 residues located within the N-terminal A/B domain of the PPARα protein, leading to an increase in ligand-dependent transactivation of targets and enhanced cooperativity with the transcriptional coactivator PGC-1. It was speculated that PPARα phosphorylation and the subsequent specific recruitment of PGC-1 favor cardiac mitochondrial FA β-oxidation during periods of stress (149). In contrast, during cardiac hypertrophic growth, the ERK-mediated phosphorylation of PPARα was shown to have an opposite effect to that of p38 kinase, resulting in a reduced capacity for myocardial lipid and energy homeostasis (150) and demonstrating therefore the importance of cellular context. In FAO hepatic-derived cells, PPARα phosphorylation level is increased by ciprofibrate, subsequently enhancing its transcriptional activity as demonstrated by a rise in ACOX gene expression (154).

Given that substrates for MAPK phosphatase 1 (also known as DUSP1) include many members of the MAPK family, including ERK1/2 and p38 MAPK, overexpression of MAPK phosphatase 1 in turn affected the constitutive PPARα activity, thus acting in a feedback control mechanism (155). Taken together, these studies hint at pivotal activating roles of MAPK on PPARα activity, thereby placing this receptor under the control of growth factors and various cellular stressors, as mentioned above.

In vivo phosphorylation studies show that the level of phosphorylated PPARα is increased by treatment with its agonist Wy-14,643 as well as the PKC activator phorbol myristoyl acetate (46). Particular modifications can increase a subset of PPAR-mediated functions; for example, phosphorylation at Ser179 and Ser230 in the PPARα hinge region by PKC leads to enhanced transactivation potential, but impairs its transrepression activity (52). Consistently, PKC inhibition impairs ligand-activated PPARα transcriptional activity and decreases PPARα ligand induction

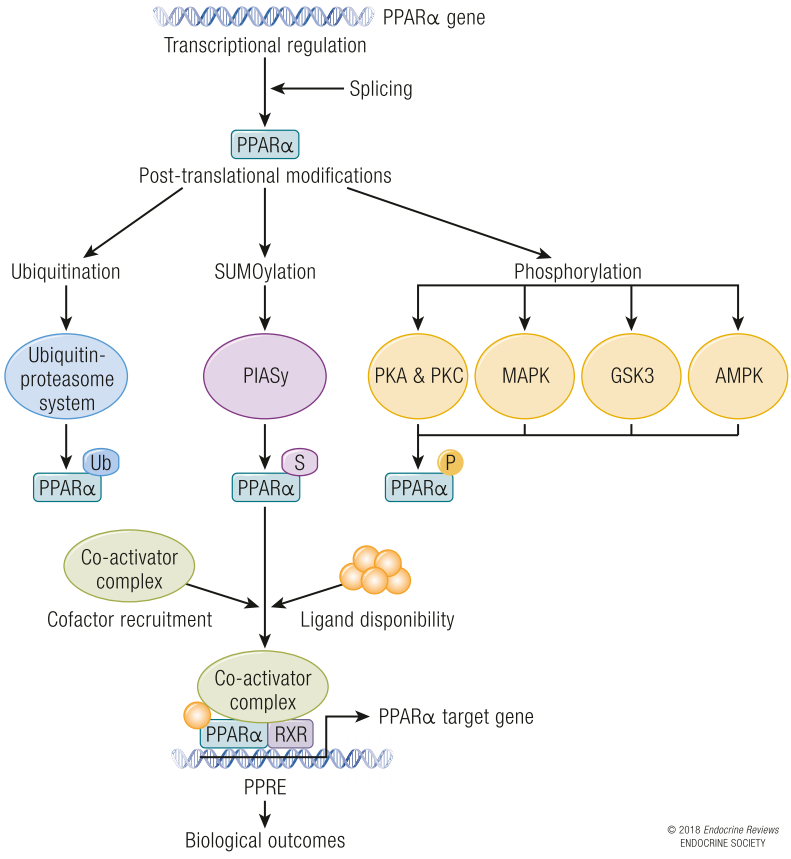


Figure 3. Levels of modulating PPARα activity. PPARα activity is controlled at several levels, including gene expression, protein translation, ligand selectivity, PTMs affecting receptor stability, and cofactor recruitment. P, phosphorylated; PIASy, protein inhibitor of activated STATy; S, SUMOylated; Ub, ubiquitinated.

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Table 2. Phosphorylation Modifications Affecting PPARα Activity

Kinase Involved	Residues Phosphorylated	Effect on Transcriptional Activity	Physiological Outcome	Refs.
ERK1/2 MAPK	Ser12 and Ser21 in A/B domain in hPPARα	Increased ligand-dependent transactivation	Cross-talk between insulin signaling and PPARα activity	(58, 148)
P38 MAPK and c-Jun N-terminal kinase	Ser6, Ser12, and Ser21 in A/B domain in mPPARα	Increased ligand-dependent transactivation	Increased cardiac mitochondrial FA β-oxidation during periods of stress	(149)
ERK-MAPK	Ser6, Ser12, and Ser21 in A/B domain in mPPARα	Decreased transactivation	Downregulation of PPARα and its target genes during cardiac myocyte hypertrophic growth	(150)
PKC	Ser179 and Ser230 in D domain in hPPARα	Increased transactivation Decreased transrepression	Molecular switch dissociating the transactivation and transrepression functions of PPARα	(52)
PKA	Mainly C domain	Increased ligand-dependent transactivation	Enhanced β-oxidation, production of glucose and ketone bodies under stress, fasting or exercise	(151)
AMPK	Not determined	Increased transactivation	Increased FAO and PPARα gene expression in skeletal muscle	(113, 152, 153)
GSK3	Predominantly at Ser73 in the A/B domain in rPPARα	Decreased PPARα stability	Links GSK3 signaling and PPARα activity to metabolic regulation	(42)

PPARα undergoes several phosphorylation modifications that markedly impact receptor function (details in the text).

of its target genes, including PPARα itself and CPT1, but enhances PPARα transrepression properties. Thus, the PKC signaling pathway acts as a molecular switch dissociating the transactivation and transrepression functions of PPARα (52). In support, it was proposed that the capacity of simvastatin to mitigate PKC action indirectly stimulates PPARα transrepression activities toward NF-κB and as such explains statin-mediated acute anti-inflammatory effects (156). As indicated earlier, PKC phosphorylation is thought to be important for PPARα and RXR interaction, as mutation of the PKC phosphorylation sites T129 and S179 in the hinge region of PPARα reduces heterodimerization (46).

The fasting-responsive kinase PKA enhances PPARα activity converging in the stimulation of PPARα target genes. It was speculated that under stress, fasting, or exercise, PPARα phosphorylation in response to PKA leads to an enhancement of β-oxidation and production of glucose and ketone bodies, which serve as fuel for muscles and brain (151). Thus, physiological cAMP modulators such as fasting, stress, or exercise potentially modulate PPARα activity. Of note, PKA-mediated PPARα phosphorylation occurs mainly in the DBD, whereas the enhancement of activity generally involves the AF-2 domain (151), suggesting an effect of phosphorylation on inter-domain communication. As another starvation-linked kinase, AMPK activity may influence the expression of PPARα and PPARα target genes in certain tissues. In the murine muscle cell line C2C12, for example, leptin stimulates PPARα gene expression and subsequent FAO through the activation of α2-subunit containing AMPK (113). Moreover, AMPK activation increases mRNA expression of PPARα and its target genes such as ACOX and CPT1 as well as PGC-1; this effect was correlated with an increased FAO in skeletal muscle (152). Additionally, AMPK and p38 MAPK were involved in the activation of PPARα by adiponectin in muscle cells (153). However, whether and which residues within PPARα may be phosphorylated remains an unresolved issue. Next to adiponectin, AMPK was also described as a modulator of PPARα expression in response to glucose levels (157, 158),

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suggesting that cellular energy levels might affect the PPARα-regulated signaling network in a ligand-independent manner. Recently, a novel mechanism was uncovered by which AMPK controls PPARα-driven gene expression in murine liver subjected to a prolonged fasting. Activated AMPK was recruited to promoters of glucose and lipid metabolism genes that are cocontrolled by PPARα and GRα (159). For the latter example in liver, it also remains to be discovered which residues of involved transcription factors, PPARα, GR, or both, may be phosphorylated by AMPK.

PPARα phosphorylation, besides affecting target gene regulation, also affects the stability and turnover of PPARα through ubiquitination (41, 160). Ubiquitination involves the conjugation of a small protein, ubiquitin, to the target protein via a sequential action of particular enzymes, which primarily leads to degradation of the receptor by the 26S proteasome. PPARα is targeted by the E3 enzyme murine double minute 2, which leads, depending on the cellular PPARα levels, to stabilization of receptor levels or degradation via the ubiquitin–proteasome pathway (161). The effect of degradation on the transcriptional activity is still under debate; Blanquart *et al.* (41) showed an inverse correlation between PPARα degradation and activity, which is in contrast to results obtained by Burns and Vanden Heuvel (42). Recently, a muscle ring finger-1–mediated monoubiquitination of cardiomyocyte PPARα was discovered that enhances nuclear export of the receptor and consequently inhibits PPARα-induced FAO in cardiac tissue (87).

GSK3 is a key enzyme of glycogen metabolism and able to inactivate glycogen synthase. Its activation phosphorylates PPARα predominantly at Ser73 in the A/B domain. This modification negatively affects PPARα stability and promotes ubiquitin proteasome-mediated degradation (42). PPARα ubiquitination may thus constitute a regulatory pathway that may contribute to the duration and the magnitude of the response to PPARα activators (41, 160). Functionally, when blocking GSK3 by biliverdin reductase A in the liver, a protective effect against steatosis is inferred, due to a preservation of hepatic PPARα activity (162).

Next to phosphorylation and ubiquitination, PPARs can be modified by SUMOylation. SUMOylation is a reversible modification that entails the covalent attachment of small ubiquitin-related modifier (SUMO) proteins via specific enzymes (163). Murine PPARα was shown to be SUMOylated on Lys358 (56). This particular SUMOylation triggers the interaction with GA-binding protein α bound to the 7α-hydroxylase cytochrome P450 7b1 promoter, resulting in specific downregulation of this gene. Hence, SUMO modification can be regarded as a positive regulator of mPPARα transrepressive activity (56). Likewise, human PPARα undergoes SUMOylation by SUMO-1 on Lys185 in the hinge region, leading to downregulation of its transactivation capacity through the selective recruitment of NCoR (53) [Table 3 (53, 56)]. Both SUMOylation sites are targeted by the E3 enzyme protein inhibitor of activated STATy (78, 163).

Finally, another means of regulating PPARα activity indirectly includes the phosphorylation of RXR and of various associated cofactors such as PGC-1α, silencing mediator for retinoid and thyroid hormone receptors, and NCoR [reviewed in Ref. (26)].

PTMs play a key role in modulating PPARα activity in (patho)physiological contexts. The modifying enzymes themselves are modulated by the cellular metabolic status and environment, thereby positioning PPARα into a very complex regulatory network. The level of complexity is even intensified knowing that PPARα forms part of a physical assembly for variable corepressor or coactivator complexes, which are also on their turn subjected to PTMs (26). Full understanding of these complex patterns of regulation requires more high-throughput technologies to characterize the PPARα interactome, its dynamic interactions with modifying enzymes, as well as the mechanisms by which these interactions modulate PPARα activity. These might point to drugs that can modulate the PPARα PTMs status with regard to metabolic disorders and inflammation.

PPARα Natural and Synthetic Ligands

PPARα natural ligands

Adoption of orphan NRs through the identification of their endogenous ligands is key toward understanding

Table 3. SUMOylation Modifications Affecting PPARα Activity

SUMO Ligase (E3) identified	Residues SUMOylated	Effect on Transcriptional Activity	Physiological Outcome	Refs.
PIASy	SUMO-1, Lys358 in the LBD region in mPPARα	Transcriptional repression	Sex-specific gene repression and protective effect in estrogen-related liver diseases, including inflammation and cholestasis	(56)
	SUMO-1, Lys185 in the hinge region in hPPARα	Downregulation of transactivation	Decrease in NCoR-specific hPPARα target gene expression	(53)

PPARα undergoes several SUMOylation modifications that markedly impact receptor function (details in the text). Abbreviation: PIASy, protein inhibitor of activated STATy.

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their biology (13). The promiscuity of PPARα for its ligands can be attributed to the large size of the LBP. PPARα is activated by a wide range of endogenous or naturally occurring biological molecules encompassing a variety of FAs and FA derivatives, including acyl-CoAs, oxidized FAs, eicosanoids, endocannabinoids, phytanic acids (164–165), and long-chain dietary FAs (166, 167) [listed in Table 4 (19, 23, 164, 168–196)]. As such, FAs and their metabolites that interact with PPARα can be derived from the diet or obtained via *de novo* synthesis. Many of those FAs exhibit an affinity for PPARα within a nanomolar range (165). This supports the fact that PPARα serves as an intracellular lipid sensor. It was proposed that PPARα senses certain endogenous lipid metabolism intermediates as ligands to regulate the expression of target genes involved in their metabolism (197). *In vivo* evidence for this came from studies using ACOX1 knockout mice. The disruption of this first and rate-limiting enzyme of the peroxisomal FAO pathway caused accumulation of long-chain fatty acyl-CoA coinciding with an increased peroxisome proliferation, hepatocarcinoma, and profound activation of PPARα in rodents (197, 198). These findings suggested that substrates of the ACOX1 enzyme may possibly be PPARα agonists. Numerous arachidonic acid metabolites serve as ligands for PPARα; in this context, 8(*S*)-hydroxyeicosatetraenoic acid (HETE), an intracellular hydroxy FA derived from the oxygenation of arachidonic acid, mainly by lipoxygenases and monooxygenases, was shown to be a strong activator of PPARα (169). Additionally, 20-HETE, a main product of cytochrome P450 (CYP4A and CYP4F) activity that catalyzes the conversion of arachidonic acid, can bind to and activate PPARα in HepG2 cells leading to enhanced expression of the PPARα target gene, apolipoprotein A-I (APOA1) (170). Leukotriene B_4 (LTB_4) is another putative PPARα ligand, which connects the receptor to inflammation and immune responses. Its ability to act as a PPARα agonist has been reported in several studies (164, 171, 199), and the binding affinity is found to be within the nanomolar range (171). Stimulation of 5-lipoxygenase activity leads to activation of PPARα via production of intracellular LTB_4 (164, 172). In endothelial cells, oxidized phospholipids were found to activate PPARα in a phospholipase A_2–dependent manner, suggesting that phospholipids may be a precursor of endogenously generated PPARα ligands (173).

Mice with liver-specific inactivation of FA synthase (FAS), a key lipogenic enzyme, developed hypoglycemia, fatty liver, and defects in expression of PPARα target genes when subject to a fat-depleted diet. These effects were reversed by a synthetic PPARα agonist, identifying thereby the *de novo* synthesized end products of FAS as endogenous activators of PPARα in adult liver (200). This phenomenon is not limited to the liver. Indeed, selective inactivation of FAS in the hypothalamus is found to affect PPARα-dependent gene expression and feeding behavior, which are corrected after hypothalamic infusion of a PPARα activator (201). These observations suggest that FAS, which is known to be regulated by nutrition, is involved in the generation of endogenous PPARα ligands. The same research group demonstrated that FAS generates 1-palmitoyl-2-oleoyl-*sn*-glycerol-3-phosphocholine, which seems to be a putative PPARα ligand (174).

Other natural compounds such as polyphenols have been identified as ligands of PPARα (175, 177). PPARα activation with resveratrol, a natural polyphenol, and some of its derivatives and analogs results in brain protection against stroke (176, 178). Resveratrol was identified as an AMPK activator, linking this sensor of the nutritional status again to PPARα functionality (202) (see "Modulating Factors of PPARα Activity"). Because PPARα acts as an intracellular sensor of FAs levels, of which an alteration is associated with the development of several diseases such as diabetes, obesity, atherosclerosis, and inflammation, finding a way of regulating the receptor activity through modulating (levels of) endogenous ligands may have the potential to generate new therapies.

PPARα synthetic ligands

Synthetic PPARα ligands include fibrates such as clofibrate, fenofibrate, and bezafibrate (listed in Table 4). Fibrates are used in the pharmacological treatment of hypertriglyceridemia (48, 203) despite the low potencies of these first-generation agonists on PPARα. Fibrates act by decreasing TG-rich lipoproteins in serum through an increase in genes involved in FAO and a decrease in apolipoprotein C-III ($APOC_3$) gene expression (204, 205). See "PPARα Agonists in a Clinical Setting" for a current status of PPARα-targeted therapies in a clinical setting.

Other synthetic compounds that are not approved for clinical use, but are often used for research purposes to activate PPARα, include GW7647, GW9578, and Wy-14,643 (also known as pirinixic acid) (29). Additionally, other chemicals with environmental impact, particularly plasticizers (*e.g.*, DEHP, used in the manufacturing of plastics) and herbicides, have been shown to function as PPARα ligands (28). As their potential impact in human health is unclear, further research is necessary to address this with respect to lifetime health.

Selective PPARα modulators

The development of a selective PPARα agonist may provide a good approach to maximize the therapeutic potential of these drug targets. Accordingly, a new generation of highly potent and selective PPARα modulators (SPPARMαs) is being developed to enable greater pharmacological effects and to dissociate the benefits of the PPARα agonists from their side effects

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Table 4. PPARα Natural and Synthetic Ligands and Their Physiological Roles

Natural Ligands			
Type	**Ligand**	**Biological and Therapeutic Outcome**	**Refs.**
Fatty acids	Arachidonic acid	PPARα binding and activation	(19, 168)
Eicosanoids	8(S)-HETE	PPARα -mediated transcription activation and induction of adipogenic differentiation of 3T3-L1 preadipocytes	(19, 169)
	20-HETE	PPARα target gene expression (APOA1) activation	(170)
	LTB_4	LTB_4 breakdown stimulation, which may contribute to the resolution of inflammation	(19, 164, 171, 172)
Phospholipids	Oxidized phospholipids	PPARα activation in a phospholipase A_2–dependent manner in endothelial cells	(173)
	Phosphatidylcholine (1-palmitoyl-2-oleoyl-*sn*-glycerol-3-phosphocholine)	Interaction with PPARα-LBD and coactivator peptide motifs	(174)
		PPARα-dependent gene induction and hepatic steatosis decreased	
Polyphenols	Resveratrol	PPARα activation in cell-based reporter assays.	(175–178)
		Brain protection against ischemic stroke in mice	

Synthetic Ligands				
Ligand	**EC_{50}**	**Status of Clinical Development**	**Biological and Therapeutic Outcome**	**Refs.**
Clofibrate	50 (m)–55 (h) μM	In clinical use	Decreased the lipid levels in hypercholesterolemic patients, mainly as the results of a reduction in VLDL, and less in LDL fraction	(179–181)
Fenofibrate	18 (m) –30 (h) μM	In clinical use	Improved the plasma lipid profiles and lipoprotein profiles	(181–183)
			Decreased atypical sphingolipids in plasma of dyslipidemic patients	
Bezafibrate	20 (m) –80 (h) μM	In clinical use	Decreased plasma TG levels and increased HDL-C levels in dyslipidemic and diabetic patients	(184)
Gemfibrozil	61 (h) μM	In clinical use	Increased HDL-C levels and lowered TG levels	(183, 185, 186)
			Coronary heart disease and stroke prevention	
WY14643	0.63 (m) –5 (h) μM	Used in basic research only	Agonist activity in functional assays	(19, 23, 168, 179, 181)
			Inhibition of NF-κB transcriptional activity and decreased inflammatory responses *in vitro* and *in vivo*	
GW9578	0.005 (m) –0.05 (h) μM	Used in basic research only	Potent lipid lowering activity	(181, 187, 188)
			Decreased LDL-C in Sprague-Dawley rats	
			Improved insulin sensitivity in obese Zucker rats	
GW7647	0.001 (m) –0.006 (h) μM	Used in basic research only	Agonist activity in GAL4-PPAR binding assay	(189)
			Decreased TGs in fat-fed hamsters and rats	

(*Continued*)

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Table 4. *Continued*

Synthetic Ligands				
Ligand	**EC_{50}**	**Status of Clinical Development**	**Biological and Therapeutic Outcome**	**Refs.**
Pemafibrate (K-877)	1 nM (h)	Phase 3 trials ongoing for treatment of dyslipidemia.	Improved lipid profile in patients with dyslipidemia Enhances RCT and decreased inflammation and atherosclerosis	(190–192)
Dual PPARα/β(δ) agonists				
Elafibranor (GFT505)	10–20 nM	Phase 3 trial ongoing for treatment of NASH	Improved hepatic steatosis, inflammation and fibrosis in NASH rodent models and in patients with NASH Improved lipid profile and insulin sensitivity in patients with dyslipidemia or prediabetes and in obese individuals	(193–196)

Abbreviations: EC_{50}, 50% effective concentration; (h), EC_{50} for human PPARα; (m), EC_{50} for murine PPARα; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein.

(192). A number of SPPARMαs have to date been identified by *in vitro* and preclinical studies and some have entered early clinical trial studies with promising results (see "PPARα Agonists in a Clinical Setting"). Among these modulators, elafibranor (GFT505), a dual PPARα/δ agonist, shows protective effects on liver steatosis, inflammation and fibrosis (196). Pemafibrate (K-877), a recently developed highly specific PPARα agonist with selective PPAR modulator (SPPARM) properties (190) shows more potent PPARα-activating efficacy *in vitro*, better effects on TGs and HDL-C levels in patients with atherogenic dyslipidemia, and a reduced risk of side effects as compared with fenofibrate. Recently, pemafibrate was shown to exert beneficial effects on lipid metabolism, reverse cholesterol transport (RCT), and inflammation, resulting in antiatherogenic properties. Therefore, pemafibrate thus seems an ideal next-generation PPARα-triggering candidate to target atherogenic dyslipidemia and cardiovascular disease risk (190).

PPARα Target Genes and Their Function

Numerous studies with PPARα-deficient mice during the past years have demonstrated the critical roles played by this receptor in energy metabolism, hepatic functions, inflammation, cardiac pathophysiology, cell cycle alterations, and rodent hepatocarcinogenesis (27). We next discuss functionalities of PPARα target genes and proteins per tissue type. Table 5 (43, 48, 71, 206–290) summarizes the many genes that have been identified as PPARα target genes in different organs since the end of the 1990s. We have indicated whether the regulation was found to be direct or indirect.

PPARα target genes in liver

The PPARα subtype exerts its main functions in the liver, where the receptor is involved in various aspects of lipid metabolism. These functions range from FA transport, binding, uptake, synthesis, degradation through mitochondrial or peroxisomal oxidation, storage, lipoprotein metabolism, and ketogenesis under fasting conditions (111). Hence, although the phenotype of PPARα-deficient mice is normal, fasting results in increased plasma-free FA (291) levels, accumulation of lipids in the liver, hypoketonemia, hypoglycemia, and hypothermia (110, 213, 225, 292). The wide range of genes that are expressed after PPARα activation indicates that this receptor has a central regulatory role in lipid metabolism (293). Recently, in a hepatocyte-specific PPARα knockout mouse model impairment in liver and whole-body FA homeostasis was observed, resulting in hepatic lipid accumulation and hypercholesterolemia during aging (294). Still today, novel mechanisms are discovered. The molecular pathway that leads to the upregulation of PPARα during fasting was recently investigated by Maus *et al.* (295). They found that FFAs bind to PLCγ-coupled cell surface receptors, such as the FA transporter CD36, in the liver and heart. This binding activates a specific pathway that leads to Ca^{2+} influx across the plasma membrane, called store-operated Ca^{2+} entry. Subsequently, increased intracellular Ca^{2+} levels induce in a cAMP-dependent manner PGC-1α and PPARα expression, ultimately leading to a transcriptional reprogramming of the cell to FA utilization (295).

Collectively, these data demonstrate that PPARα is a master regulator of FA catabolism by governing the regulation of key enzymes involved in peroxisomal, mitochondrial, and microsomal FAO.

Additionally, PPARα also plays an important role in ketogenesis by regulating expression of HMG-CoA

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Table 5. List of Selected PPARα Target Genes

Target Gene	Official Gene Symbol	Gene Functions	Comments	Fibrates WT	Fibrates KO	Refs.
Liver						
FA transport protein (FATP)	*SLC27A1*	Facilitates the uptake of long-chain FAs by the liver Fatty acid transport	Direct PPARα target with a functional PPRE in its promoter	↑	=	(206–208)
Fatty acid translocase (FAT/CD36)	*CD36*	Uptake of long-chain FAs and oxidized LDL	Transcriptional mechanism remains unclear; induction goes through the distal promoter region of the gene even though no PPRE could be identified	↑	=	(208–210)
Acyl-CoA synthetase	*ACS*	Catalyzes FA metabolism by converting inactive FAs into active acyl CoA derivatives	Direct PPARα target with a functional PPRE in its promoter	↑	=	(207, 211–214)
Fatty acid binding protein	*FABP*	Intracellular lipid trafficking	Functional PPRE was identified; FABP2 and FABP1 have different effects on the ability of different agonists to promote PPARα activation ligand dependently	↑	=	(215–217)
Acyl-CoA binding protein (ACBP)	*DBI*	Fatty acyl CoA esters transport	Functional PPRE was identified; ACBP expression is regulated by fasting/refeeding in rodents; a dual regulation by PPARα and SREBP-1c might be considered.	↑	=	(218, 219)
Carnitine palmitoyl-transferase 1	*CPT1A*	Catalyzes the transfer of a long-chain fatty acyl group from CoA to carnitine	Functional PPRE was identified	↑	=	(220)
Carnitine palmitoyl-transferase 2	*CPT2*	Conjugates the FA back to CoA for subsequent β-oxidation	Functional PPRE was identified	↑	=	(213, 221)
Medium-chain acyl-CoA dehydrogenase (MCAD)	*ACADM*	Mitochondrial FA β-oxidation	Functional PPRE was identified	↑	=	(222–225)
Long-chain acyl-CoA dehydrogenase (LCAD)	*ACADL*	Mitochondrial FA β-oxidation	No PPRE /unknown mechanism	↑	=	(213, 226)
Very long chain acyl-CoA dehydrogenase (VLCAD)	*ACADVL*	Mitochondrial FA β-oxidation	No PPRE reported	↑	=	(213, 227, 228)
Dodecenoyl-CoA δ-isomerase	*ECI1*	Mitochondrial β-oxidation of unsaturated and saturated FAs	Using microarray, expression of ECI1 was found to be decreased in PPARα-deficient mice but no PPRE has yet been identified	↑	?	(229, 230)
Acyl-CoA oxidase	*ACOX*	Peroxisomal β-oxidation	Functional PPRE was identified; the fasting-induced activation of ACOX expression was markedly blunted in PPARα-deficient mice	↑	=	(71, 164, 225, 229, 231)
Bifunctional enzyme (BIEN)	*EHHADH*	Peroxisomal β-oxidation	Three different PPREs have been identified in the promoter sequence of the BIEN gene, yet only one allows for efficient transactivation of the BIEN gene by PPARα	↑	=	(216, 229, 231, 232)

(*Continued*)

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Table 5. *Continued*

Target Gene	Official Gene Symbol	Gene Functions	Comments	Fibrates WT	Fibrates KO	Refs.
Peroxisomal 3-ketoacyl-CoA thiolase	*ACAA1*	Peroxisomal β-oxidation	It is unclear whether the DR-1 sequence identified within its promoter mediates PPARα transactivation; differential gene regulation in human vs rodent hepatocytes	↑	=	(71, 232–234)
Peroxisomal membrane protein 11A PEX11α	*PEX11A*	Peroxisome proliferation	A functional PPRE 8.4 kb downstream of the PEX11α was identified by a gene transfection study	↑	?	(235–237)
Cytochrome P450 4A	*CYP4A*	Microsomal FA ω-hydroxylation	In rodents, CYP4A1 upregulation through a functional PPRE; in humans, no PPRE detected in the 5′-flanking region of the CYP4A11 gene; induction upon fasting or fenofibrate treatment in transgenic mice	↑	=	(229, 238–240)
Mitochondrial 3-hydroxy-3-methylglutaryl-CoA synthase (mHMG-CoAS)	*HMGCS2*	Ketogenesis/ condensation of acetyl-CoA and acetoacetyl-CoA to generate HMG-CoA	Functional PPRE identified in the rat and pig mHMG-CoAS gene; mHMG-CoAS proteins interact physically with PPARα and act as a coactivator to upregulate transcription from the PPRE of its own gene	↑	=	(229, 241)
Fibroblast growth factor 21	*FGF21*	Metabolic fuel homeostasis during ketosis	Functional PPRE identified; induction upon fasting and PPARα agonists treatment	↑	=	(242–244)
FA desaturase 2 (Fads2)	*FADS*	Lipogenesis	DR-1 that functions in the regulation of human FADS2 gene transcription identified; an indirect regulatory mechanism involving SREBP-1c might be considered	↑	=	(245–250)
Stearoyl-CoA desaturase (Scd1)	*SCD*	Lipogenesis	Functional PPRE identified; an indirect regulatory mechanism involving SREBP-1c might be considered	↑	=	(245–247)
Malic enzyme (Mod1)	*ME1*	Lipogenesis	Functional PPRE identified; decrease of ME1 expression in livers of PPARα-deficient mice	↑	=	(210, 229, 251)
Phosphatidate phosphatase (Lpin2)	*LPIN2*	Lipogenesis	Dual target of PPARα and PPARβ/δ; induction by fasting and dietary FA maintained in PPARα-deficient mice	↑	=	(252, 253)
Acetyl-CoA carboxylase (ACC)	*ACACA*	Lipogenesis	PPARα might indirectly influence ACC and FAS expression by controlling metabolic processes	↑	?	(248, 254, 255)
Fatty acid synthase (FAS)	*FASN*	Lipogenesis	In human primary hepatocytes, PPARα agonists, cooperatively with insulin and LXR agonists, induce FAS and ACC	↑	?	
Liver X receptor α (LXRα)	*NR1H3*	Cholesterol metabolism	Induction in macrophages upon PPARα treatment; decreased in livers upon fasting in PPARα-deficient mice	↑	?	(256, 257)

(*Continued*)

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Table 5. *Continued*

Target Gene	Official Gene Symbol	Gene Functions	Comments	Fibrates WT	Fibrates KO	Refs.
Cholesterol 7α-hydroxylase (Cyp7a1)	*CYP7A*	Bile acid metabolism	Functional PPRE was identified; murine Cyp7A1 species, harbors an additional PPRE; indirect mechanism involving HNF4α transactivation activity was also reported	#	?	(258–261)
Cholesterol 27α-hydroxylase (Cyp27a1)	*CYP27A1*	Bile acid metabolism	PPARα affects the stability of Cyp27a1 mRNA; in rodents, fibrates suppress bile acid biosynthesis via PPARα-mediated downregulation of Cyp27a	↓	=	(257, 261)
Sterol-12a-hydroxylase	*CYP8B1*	Bile acid metabolism	Functional PPRE identified	↑	=	(262)
Phosphoenolpyruvate carboxykinase (Pck1)	*PKC1*	Gluconeogenesis	Elusive regulation mechanism, although a functional PPRE was shown to be functional in adipocytes; PPARα-mediated Pck1 regulation only observed in human hepatocytes; induction during fasting reported	?	?	(252, 263, 264)
Glycerol-3-phosphate dehydrogenase (GPDH)	*GDP*	Metabolic conversion of glycerol into glucose	PPARα directly governs glycerol metabolic by directly upregulating expression of these genes; upregulation by fasting only in wild-type mice	↑	=	(264)
Glycerol kinase,	*GK*					
Glycerol transporters aquaporins 3 and 9	*AQA3, AQA9*					
Pyruvate dehydrogenase kinase isoform 4 (Pdk4)	*PDK4*	Glucose oxidation	No PPRE yet identified; in wild-type PPARα mice, PDK4 expression increased in response to fasting with suppression by refeeding	?	?	(265, 266)
Glycogen synthase 2 (Gys-2)	*GYS2*	Glycogen synthesis	Functional PPRE identified; regulation mechanism involving competition between HNF4α and PPARα reported; Gys-2 mRNA reduction is PPARα-deficient mice and in fasted and refed state	↑	=	(267)
Lipoprotein lipase	*LPL*	Hydrolysis of TGs	Functional PPRE identified	↑	=	(48, 208, 230, 268)
Angiopoietin-like protein 4	*ANGPTL4*	Inhibitor of LPL activity	Increased upon fasting	↑	=	(269, 270)
Rev-erbα	*NR1D1*	Repressor of gene transcription	Functional PPRE identified; negative interference with the activation of APOC3 gene promoter by HNF4	↑	?	(47, 268, 271)
Apolipoprotein AI	*APOA1*	Plasma HDL metabolism	In humans, a functional PPRE was identified; in rodents, an indirect mechanism involving the PPARα-dependent induction of the Rev-erbα was reported	↓	=	(271–273)
Apolipoprotein AII	*APOA2*	Plasma HDL metabolism	Functional PPRE identified; induced APOA2 plasma levels and hepatic mRNA expression in humans; both a marginal reduction and induction upon fibrate treatment observed in rodents	↑	=	(274–277)
Apolipoprotein A-V	*APOA5*	Plasma TG metabolism	Functional PPRE identified	↑	?	(278, 279)

(*Continued*)

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Table 5. *Continued*

Target Gene	Official Gene Symbol	Gene Functions	Comments	Fibrates WT	Fibrates KO	Refs.
Apolipoprotein C-III	*APOC3*	Plasma HDL Metabolism	Transcriptional suppression by fibrates involves HNF4	↓	=	(280–282)
Heart						
Acyl-CoA oxidase	*ACOX*	Peroxisomal β-oxidation	Reduced fasting-induced ACOX expression in PPARα-deficient mice	?	?	(71, 225)
Carnitine palmitoyl-transferase 1	*CPT1*	Fatty acid transport/oxidation	Functional PPRE identified; reduced fasting-induced CPT1 expression in PPARα-deficient mice	↑	?	(225, 283, 284)
Carnitine palmitoyl-transferase 2 (CPT2)	*CPT2*	Fatty acid oxidation	Functional PPRE identified	?	?	(221, 285)
Long-chain acyl-CoA synthetase (LCAS)	*ACADL*	Fatty acid activation	Functional PPRE identified	?	?	(211, 285)
Medium-chain acyl-CoA dehydrogenase (MCAD)	*ACADM*	Mitochondrial FA β-oxidation	The fasting-induced activation ACADM expression was abolished in PPARα-deficient mice	?	?	(223, 225, 285)
Muscle						
Carnitine palmitoyl-transferase 1	*CPT1*	Fatty acid transport/oxidation	Functional PPRE identified	↑	?	(284, 286)
Pyruvate dehydrogenase kinase isoform 4 (Pdk4) (soleus)	*PDK4*	Glucose oxidation	No PPRE has yet been identified	↑	?	(284, 287)
White adipose tissue						
Fatty acid translocase (FAT/CD36)	*CD36*	Uptake of long-chain FAs and oxidized LDL	Transcriptional mechanism remains unclear	↑	↑	(288)
Acyl-CoA oxidase	*ACOX*	Peroxisomal β-oxidation	Functional PPRE identified	↑	↑	(71, 288)
Lipoprotein lipase	*LPL*	Hydrolysis of TGs	Functional PPRE identified; basal levels of LPL mRNA were higher in PPARα-deficient mice as compared with wild-type mice	↑	=	(268, 288)
Glycogen synthase 2 (Gys-2)	*GYS2*	Glycogen synthesis	Gys-2 mRNA is reduced in PPARα-deficient mice	?	?	(267)
Small intestine						
Pyruvate dehydrogenase kinase isoform 4 (Pdk4)	*PDK4*	Glucose oxidation	Intestine-specific regulation of PDK gene transcription by LXR	?	?	(289)
Carnitine palmitoyl-transferase 1	*CPT1*	FA transport/oxidation	Intestine-specific regulation of CPT1 gene transcription by LXR	?	?	(289)
Apolipoprotein C-III	*APOC3*	Plasma HDL metabolism	PPARα-dependent gene repression by fibrates *in vivo*	↓	?	(288, 290)
Fatty acid binding protein	*FABP*	Intracellular lipid trafficking	Reduced FABP mRNA in PPARα-deficient mice	↑	?	(215, 288)

PPARα regulates the expression of its target genes involved in different biological process via distinct mechanisms (more details for some examples are discussed in the text). ↑ represents an increase upon fibrate treatment; ↓ represents a decrease upon fibrate treatment; = represents no change upon fibrate treatment; ? represents no information available on a change; # represents conflicting results.
Abbreviations: KO, knockout PPARα mouse; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein; WT, wild-type mouse.

synthase 2, a rate-limiting enzyme of ketogenesis, which catalyzes condensation of acetyl-CoA and acetoacetyl-CoA to generate HMG-CoA (296). Accordingly, the expression of mHMG-CoA synthase gene, harboring a functional PPRE (296) in its promoter, is impaired in PPARα-null mouse liver (241).

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Of note, HMG-CoA synthase specifically stimulates its own transcription rate by acting as a coactivator of PPARα. A recent report identified liver-specific PPARα as a critical factor for ketogenesis. It was shown that PPARα protects from fish oil–induced oxidative stress leading to acute liver failure through its regulatory actions on ketone body levels, hence suggesting a potential therapeutic advantage of ketone bodies such as β-hydroxybutyrate in preventing lipid peroxidation-induced liver damage (297). Of particular importance in regulating ketogenesis, in addition to FAO, is the metabolic fibroblast growth factor 21 (FGF21). Its hepatic expression is PPARα-dependent and is induced by fasting, a ketogenic diet, and fenofibrate (242). Moreover, upon fasting, FGF21 production required cAMP-responsive element-binding protein H (CREBH), the liver-enriched transcription factor. Interestingly, it was reported that CREBH and PPARα act as interactive *trans*-activators that regulate each other's expression and act in synergy to activate the expression of their target gene FGF21 to regulate lipolysis, FAO and ketogenesis upon fasting or under an atherogenic HFD (298). Of note, CREBH acetylation is required for the fasting-induced CREBH-PPARα interaction *in vivo* (299). These findings on synergistic cross-talk between PPARα and CREBH might provide a regulatory mechanism for NRs or transcription factors to extend their biological actions and thereby coordinate adaptive physiological responses.

As far as lipogenesis is concerned, besides examples listed in Table 5, Oosterveer *et al.* (248) showed that induction of some lipogenic genes upon fibrate treatment is regulated transcriptionally by SREBP-1c. Consistently, in SREBP-1$^{-/-}$ mice, chronic PPARα activation with fenofibrate failed to induce lipogenic genes such as stearoyl-CoA desaturase and FA desaturase 2, suggesting an indirect regulatory mechanism (248). Concordantly, in primary hepatocytes, the PPARα agonists Wy-14,643 and GW7647 increased SREBP-1c activity by triggering PPARα activity on a DR-1 element in the human SREBP-1c promoter (255). Moreover, in PPARα-deficient mice, a disturbance in the normal regulation of SREBP-1–sensitive lipogenic genes such as acetyl-CoA carboxylase and FAS was observed (254). The effect of PPARα agonists on SREBP targets has been ascribed to increased activation of SREBP-1c via enhanced proteolytic cleavage (214). PPARα agonists increase the protein levels of the mature form of SREBP-1c by increasing the rate of proteolytic cleavage of its membrane-bound precursor form, without changing its mRNA levels. Of note, PPARα can also indirectly modulate SREBP-1c transcription via cross-regulation with LXRα signaling pathway. In this line, Hebbachi *et al.* (300) reported that the intact PPARα is required to mediate the LXRα-dependent response of Scd-1 and FAS gene expression to insulin and that an exogenous nonsteroidal synthetic ligand of LXRα, TO-901317, was effective in rescuing PPARα deficiency, suggesting participation of PPARα in generating endogenous LXRα ligands (300).

Additionally, malonyl-CoA decarboxylase (MCD) is transcriptionally upregulated following PPARα activation. MCD degrades the lipogenic intermediate malonyl-CoA, which is both a precursor of FA biosynthesis and an inhibitor of the mitochondrial FA transporter CPT-1 (301). PPARα hence might inhibit the lipogenic pathway indirectly by inducing MCD, in line with its role in promoting energy expenditure (302, 303).

Evidence exists for a functional role of PPARα in bile acid homeostasis; however, the mechanism is not understood. Although various studies have reported on the expression modulation of Cyp7A1, a microsomal cytochrome P450 isoenzyme that regulates *de novo* hepatic biosynthesis of bile acids by PPARα, contradictory observations were made (listed in Table 5). In transfected cells, both murine and human cholesterol 7α-hydroxylase (Cyp7a1) gene promoters were stimulated by FA or Wy-14,643 treatments; however, the murine promoter harboring an additional PPRE displayed a greater activity (258). Additionally, the expression level of Cyp7a1 diminished in PPARα-deficient mice, thus revealing the involvement of PPARα (262). In contrast, bezafibrate treatment decreases liver bile acids and Cyp7a1 expression in wild-type mice and PPARα-deficient mice (304). Moreover, a Cyp7a1 decrease was observed in wild-type mice upon ciprofibrate treatment but not in PPARα-deficient mice (261). Besides, other genes involved in bile acid synthesis are regulated by PPARα, including cholesterol 27α-hydroxylase, whose activity was downregulated by ciprofibrate treatment in a PPARα-dependent manner (261), and sterol-12a-hydroxylase (rat Cyp8B1), which was reported to be directly upregulated by PPARα through a functional PPRE (262).

PPARα-deficient mice display severe hypoglycemia after fasting (110), illuminating the central role of PPARα in glucose metabolism. Plasma glucose is maintained during fasting by combination processes including *de novo* glucose synthesis (gluconeogenesis), glycogen breakdown (glycogenolysis), and control of peripheral glucose utilization. Fasting-induced hypoglycemia could be caused by inhibition of either of these pathways. Interestingly, genes involved in gluconeogenesis have been identified as PPARα targets, including phosphoenolpyruvate carboxykinase (Pck1), pyruvate carboxylase, and lactate dehydrogenase A (252). Although it was originally proposed that Pck1 may be specifically regulated by PPARα in human but not mouse hepatocytes (264), PPARα was shown to activate Pck1 promoter activity in murine hepatoma Hepa1c1c7 cells (305), whereas in the livers of rats treated with clofibrate a reduced expression level of Pck1 was observed (306). Additionally, PPARα

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promotes the metabolic conversion of glycerol into glucose through the induction of several hepatic enzymes such as glycerol-3-phosphate dehydrogenase and glycerol kinase, and of glycerol transporters aquaporins 3 and 9; all of these genes are identified as targets of PPARα (264). Moreover, PPARα contributes to gluconeogenesis by inducing the expression of pyruvate dehydrogenase kinase isoform 4 (307). Pyruvate dehydrogenase kinase isoform 4 phosphorylates and thereby inactivates the pyruvate dehydrogenase complex, resulting in a conservation of pyruvate as gluconeogenic substrate (308). Besides, PPARα targets glycogen synthesis in liver by regulating its direct target, glycogen synthase 2, which is a rate-limiting enzyme in the storage of glycogen in liver and adipose tissue. Consistent with this, Wy-14,643 was shown to stimulate glycogen synthase 2 expression in wild-type but not in PPARα$^{-/-}$ hepatocytes (267). TRB3, another PPARα-target gene, is induced in liver during fasting and in diabetic mice in a PGC-1–dependent manner. Upregulation of TRB3 expression through direct transcriptional control by PPARα inhibits insulin signaling by binding to and preventing activation of Akt/PKB. The latter kinase is a positive regulator of the cellular response to insulin (309, 310). How this can be reconciled with the profile of PPARα agonists in other models or clinically, in humans, is as yet unclear.

PPARα has also a profound impact on amino acid metabolism; specifically, it downregulates genes involved in major pathways of amino acid metabolism, including transamination, deamination, urea cycle, oxidation of α keto acids, and synthesis of amino acid–derived products. In support of a role of PPARα in amino acid gene expression, plasma urea levels are elevated in PPARα-deficient mice. The molecular mechanisms underlying downregulation of amino acid metabolism by PPARα remain elusive; however, some studies proposed that PPARα may modulate the activity of other transcription factors that are directly involved in amino acid homeostasis, including HNF4α and CCAAT/enhancer-binding protein (C/EBP)α (311).

PPARα target genes in heart

During the transition from fetal to adult cardiomyocytes, both PPARα and PGC-1α expression are increased, corresponding with the primary use of FAO as an energy source (312). In the context of a pathological cardiac hypertrophy, the expression of PPARα itself is also downregulated (150, 313). Conversely, PPARα activity is increased in the diabetic heart, which is characterized by increased lipid oxidation. Correspondingly, overexpression of PPARα leads to a cardiomyopathy mimicking a diabetic heart (314–316). To dissect the *in vivo* role of PPARα in myocardial metabolism, transgenic mice with cardiac-specific expression of PPARα were generated (315, 317). Interestingly, myosin heavy chain (MHC)–PPARα mice, which developed hepatic insulin resistance, also displayed enhanced IL-6 levels in the circulation (315). Consistent with the metabolic phenotype of myocardial lipid accumulation in MHC-PPARα mice, the expression of genes involved in cellular FA transport was activated by PPARα. Conversely, cardiac glucose transport and glycolytic genes were repressed in MHC-PPARα mice (317), likely occurring via a transcriptional mechanism, as fenofibrate also downregulated GLUT4 promoter–driven luciferase reporter gene expression upon cotransfection with PPARα expression plasmid into rat neonatal ventricular cardiomyocytes (317). As mentioned also above, Oka *et al.* (81) have recently demonstrated that SIRT1, which plays an important role in metabolic adaptation to starvation conditions, suppresses FAO in failing hearts possibly by inhibiting the heterodimerization of PPARα and RXRα, suggesting thereby that preventing PPARα–SIRT1 or restoring PPARα–RXR heterodimerization may improve the progression of heart failure through modulation of metabolic remodeling (81). Taken together, these findings highlight the complexity of a balanced cardiac energy metabolism and the necessity of distinguishing the etiology of heart failure to be able to predict the outcome of PPARα agonists.

"Fibrates are generally well-tolerated drugs and have a good safety profile irrespective of concomitant therapy."

PPARα target genes in lipid metabolism

PPARα agonists have a pronounced effect on lipoprotein metabolism, which is involved in lipid and cholesterol transport. In this regard, PPARα ligands reduce very-low-density lipoprotein (VLDL) production and enhance the catabolism of TG-rich particles, which indirectly decreases small dense low-density lipoprotein (LDL) particles and enhance the formation of HDL particles and hepatic elimination of excess cholesterol (27). More specifically, PPARα activators decrease plasma TG-rich particle levels by increasing the activity of LPL (48, 318), which is the key enzyme in the hydrolysis of TGs. PPARα controls LPL gene expression directly through a functional PPRE located in its promoter region (268), and indirectly by reducing the levels of APOC3, a natural inhibitor of LPL activity (281), likely via a mechanism involving the transcription factors Rev-erbα, HNF4α, or forkhead box protein O1 (FOXO1) (280, 290, 319). Rev-erbα, itself induced by fibrates via a PPRE in the Rev-erbα gene promoter, negatively interferes with the HNF4-mediated activation of APOC3 gene promoter (280). FOXO1 stimulates APOC3 expression via specific binding to its target site (CCAAACA) in the APOC3 promoter. The association of FOXO1 with the APOC3 promoter was significantly reduced in response to PPARα activation, in line with the inhibition of APOC3 promoter activity in PPARα-expressing HepG2 cells (319). Besides APOC3, LPL activity

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is tightly modulated by other factors such as apolipoprotein A-V (APOA5), angiopoietin-like protein 3, and angiopoietin-like protein 4. Concordantly, PPARα increases expression levels of APOA5, which is critical for sustaining normal plasma TG levels, through functional PPREs located in its promoter (278, 279). Moreover, PPARα upregulates the expression levels of angiopoietin-like protein 4 (269), which acts as an inhibitor of LPL activity by converting active LPL dimers to inactive monomers (320), which may play a role in tissue-specific directing of FA. PPARα activators affect also the metabolism of HDL by regulating the expression of its major apolipoproteins, A-I (APOA1) and A-II (APOA2). In humans, it increases the expression of these two genes by binding to PPREs in their promoters, contributing to increased plasma HDL levels (273, 274). In rodents, however, gene expression of APOA1 is not promoted by PPARα because of disabling mutations within the PPRE. In fact, the murine APOA1 gene is even negatively regulated by PPARα agonists through an indirect mechanism involving the PPARα-dependent induction of the NR Rev-erbα, a negative regulator of transcription (271).

PPARα target gene expression in immune cells

The functional expression of PPARα by several immune cell types suggests that this receptor may play an important role in regulation of immune responses. The exact role of PPARα in, for instance, innate host defense mechanisms is however largely unknown. PPARα was shown to be essential for antimycobacterial responses in murine bone marrow–derived macrophage cells, involving gene activation of transcription factor EB and induction of lipid catabolism (321). PPARα deficiency resulted in an increased bacterial load and exaggerated inflammatory responses during mycobacterial infection (321). In human macrophages, PPARα activation upregulates the expression of ABCA1, a member of the ATP-binding cassette transporter family, by an indirect mechanism involving the induction of LXRα expression. The increase in ABCA1 expression promotes APOA1-mediated cholesterol efflux (322). Adiponectin receptor 2, an LXR target gene, is positively regulated by PPARα both in primary and THP-1–differentiated macrophages (323). Given that adiponectin receptor 2 is expressed in human atherosclerotic lesions and macrophages, modulating the expression of this gene by PPARα and LXRα may provide beneficial effects in the treatment of cardiovascular disease. This finding highlights a mechanism of cross-talk between adiponectin and PPARα–LXRα signaling pathways (323). In murine macrophages, PPARα activation was also shown to upregulate the expression of the classic PPARα target gene CPT1 (156). Furthermore, in mice, PPARα activation in hepatic Küpffer cells or monocyte-recruited macrophages results in anti-inflammatory effects via the downregulation of IL-15 and IL-18 (324).

PPARα targets linked to the rodent-only hepatocarcinoma side effect

Sustained induction of the peroxisomal FAO system accounts for the massive proliferation of peroxisomes, which may contribute to hepatocarcinoma, a property intrinsic to mouse and rat PPARα but not observed in humans (25, 27). By using hepatocyte-specific PPARα-deficient mice, PPARα agonist treatment was shown to induce hepatomegaly via hepatocyte PPARα activation. Interestingly, the proliferative response was mediated exclusively by PPARα activation in hepatocytes (324).

Several mechanisms have been proposed to elucidate the species difference in response to peroxisome proliferators. For example, in humans, lower expression levels of PPARα or the presence of splice variants was suggested to contribute to the resistance of humans to peroxisome proliferation upon treatment with fibrate drugs (50, 108). However, most of the proposed mechanisms have been met with reasonable criticism [reviewed in Ref. (325)]. The work of Cheung *et al.* (326) with "humanized" PPARα mice suggests that there are inherent species differences in the receptor that may be more critical. In this regard, PPARα-humanized mice that express human PPARα in a mouse PPARα-null background exhibit, unlike wild-type mice, a diminished hepatocellular proliferation whereas the induction of mitochondrial and peroxisomal FAO enzymes was comparable to wild-type (326). Moreover, it was proposed that the mechanism by which peroxisome proliferators lead to mitogenic and hepatocarcinogenic effects in mouse liver is due to murine PPARα-dependent downregulation of let7c microRNA, which in turn releases the repression of c-myc, an important oncogene that is related to liver cancer development. Interestingly, this effect does not occur in the humanized PPARα mice (327); thus, the induced expression of c-myc protein through species-specific regulation of a microRNA contributes to PPARα ligand–induced liver hyperplasia in mice (327). Still, some questions arise regarding the fact that the PPARα-humanized mouse does not precisely model humans because there could be species differences in coactivators, corepressor, and target gene response elements.

PPARα Agonists in a Clinical Setting

PPARα is widely expressed and serves to activate FAO pathways in the liver in response to fasting. This retunes energy metabolism to permit efficient use of mobilized lipids in the fasted state. The resulting changes may lower secretion of TGs from the liver and promote TG clearance from plasma, and thereby also reduce atherogenic lipoprotein particles. It was this activity that prompted therapeutic development to reduce cardiovascular risk in patients with raised plasma lipids. In clinical trials, a marked reduction in

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circulating TGs with a rise in HDL-C was consistently documented, with reductions in LDL cholesterol (LDL-C) depending on the pretreatment TG levels [reviewed in Ref. (27)].

By virtue of their TG-lowering capacity, the official indication for fibrates is hypertriglyceridemia. For hypertriglyceridemia, fibrates are the first choice when a change of diet is not sufficient. For dyslipidemia and hypercholesterolemia, statins remain the first choice because of a higher potency in lowering LDL-C (328).

Fibrates are generally well-tolerated drugs and have a good safety profile irrespective of concomitant therapy (329). However, changes in biochemical markers, some of which related to pathological states, may occur. Among these are elevations in homocysteine, creatinine levels, alanine aminotransferase, and aspartate aminotransferase. Of note, lowering of homocysteine with folic acid did not affect cardiovascular diseases (330, 331). Along the same line, even after prolonged treatment, the increase of creatinine is rapidly reversible upon treatment cessation, indicating absence of irreversible tissue damage (332). Muscle-related adverse effects, such as an increase in creatine kinase and myalgia, may occur but rhabdomyolysis has only been observed upon combining gemfibrozil and cerivastatin, an effect due to classical P450-mediated drug interaction. Rhabdomyolysis was found to be rare with (single) fibrate therapies (333).

PPARα agonists and effects on lipid and lipoprotein metabolism in humans

The prominent role of PPARα in governing lipoprotein and lipid metabolism in humans was demonstrated by the fact that fibrates, widely prescribed drugs to treat dyslipidemia, act by binding to PPARα and induce PPARα-dependent gene transcription (6, 334).

A plethora of clinical studies have shown that fibrate treatment lowers serum TG concentration by ~30% [−35%, $P < 0.01$ (335); −30%, $P < 0.045$ (336); −40% to −20%, $P < 0.001$ (337); −37% (338); −38% (328)], reduces VLDL TG by ~40% [−41%, $P < 0.01$ (335); −44% to −20%, $P < 0.001$ (337)], and reduces (in combination with statins) the atherogenic small dense LDL by ~20% [−23% (328)]. Additionally, HDL increases slightly [+19%, $P < 0.02$ (335); +82% to +106%, $P < 0.005$ (337); +14% (328)] and serum concentrations of APOA1 and APOA2 increase by ~20% [+19%, $P < 0.02$ (274, 335)]. In obese patients with type 2 diabetes, fenofibrate decreased serum TG concentrations, whereas glycated hemoglobin increased (339). In contrast, the Fenofibrate Intervention and Event Lowering in Diabetes (FIELD) and Action to Control Cardiovascular Risk in Diabetes (ACCORD) studies show glucose-neutral effects of fenofibrate, supporting the notion that the first generations of fibrates are not antidiabetic drugs (329, 332). Concordantly, insulin sensitivity was not affected by fenofibrate administration (336, 339). Nevertheless, yet controversial, a more recent study reported that fenofibrate may increase insulin sensitivity by inhibiting adipose tissue expression of retinol-binding protein-4, an adipocytokine (340). For LPL, in patients on gemfibrozil (6 weeks), there was no significant change in LPL in adipose tissue (341), possibly due to pharmacokinetic effects. Nevertheless, plasma TG levels decreased by 46% and several patients demonstrated increases in muscle LPL activity, although these were inconsistent and not statistically significant (341). Previous studies using other PPARα agonists did report a significant increase of skeletal muscle LPL activity [+39%, $P < 0.05$ (335); +50%, $P < 0.005$ (338)] and even adipose tissue LPL activity (342). It was originally proposed that gemfibrozil predominantly decreases TG levels by decreasing hepatic VLDL secretion (336, 341, 343). However increased lipolysis may rather be caused by a primary liver effect (*e.g.*, APOC3 decreased and APOA5 increased). Additionally, although fenofibrate reduces plasma TG levels, it does not seem to lower concentrations or turnover rates of nonesterified FAs in adipose tissue (336), correlating with a lack of effect on insulin sensitivity.

"Little is known about PPARα-mediated modulation of kinase activities in terms of an anti-inflammatory action."

PPARα agonists and effects on liver in humans

Dyslipidemia and insulin resistance are known risk factors for NAFLD. As PPARα activation reduced hepatic steatosis in mice, applicability of fibrates was suggested as a potential treatment of NAFLD in humans (111). In a relatively small (16 patients) and phenotypically heterogeneous cohort study, patients completed 48 weeks of therapy with fenofibrate without adverse events (344). Significant decreases in TG, glucose, alkaline phosphatase, and γ-glutamyl transpeptidase, as well as an increase of APOA1 levels, were found, yet the effects on fibrosis as addressed by liver histology were minimal (344). Similar studies with fenofibrate for 3 months confirmed results of a lowered TG yet unchanged hepatic fat content (345). Short-term treatment (2 to 8 weeks) with bezafibrate combined with diet and exercise of steatosis-suffering donors for liver transplantation was found to decrease macrovesicular steatosis (346), although in this particular study the impact of bezafibrate alone was not addressed. Treatment of nonalcoholic steatohepatitis (NASH) (347) patients with gemfibrozil (for 4 weeks) lowers alanine aminotransferase, aspartate aminotransferase, and γ-glutamyl transpeptidase plasma levels (348), but not in a pilot study with clofibrate (for 12 months) (349). The relatively weak potency of these fibrates combined with lower expression level of PPARα in human as compared with mouse liver was presented as an explanation (350). However, in all studies, plasma TG levels effectively decreased. Novel PPARα agonists with greater potency and efficacy,

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such as pemafibrate, or with dual (elafibranor) or triple (lanifibranor) agonism, may prove to be more useful in the treatment of NASH/NAFLD (351). As mentioned also in "PPARα targets linked to the rodent-only hepatocarcinoma side effect" above, it is important to emphasize that humans are resistant to the induction of neoplasms by fibrate drugs in contrast to rodents (25, 352). This implies that following ligand activation, the PPARα-mediated pathways controlling lipid metabolism work independently from those controlling cell proliferation pathways (326).

PPARα agonists and effects on the cardiovascular system in humans

As in rodents, PPARα is downregulated in the failing human heart (353). PPARα protein levels were compared between homogenates from left ventricular biopsies from five control donor hearts vs biopsies from five patients with compensated end-stage heart failure at the time of transplantation (353). Whether augmenting PPARα activity would improve the progression of heart failure is an interesting, but to date unanswered, question. Subgroup analysis of the Helsinki Heart Study (HHS) trial suggested potential benefit in preventing heart failure with gemfibrozil compared with placebo (13.3% vs 10.6%; hazard ratio, 0.76; $P = 0.04$) (354). Potential mechanisms may relate to direct cardiac and/or on PPARα's lipid lowering effects. PPARα agonists may also improve cardiac performance by decreased transcription of proinflammatory cytokines as shown *in vitro* (355) and/or as in rodent studies, by reduced activation of aldosterone-mediated myocardial apoptosis (356) and/or by decreased endothelin-1–mediated cardiac fibrosis (357). Further research is needed to assess whether fibrates, and especially the novel SPPARMs (see "SPPARMαs in humans" below), can prevent or ameliorate chronic heart failure in humans.

Gemfibrozil was shown to decrease the incidence of coronary heart disease in both the HHS and Veterans Affairs HDL Intervention Trial (VA-HIT). Treatment was concomitant with a marked increase in HDL-C and reductions in serum levels of total, LDL, and non–HDL-C and TGs (186, 354), yet there is no definite proof that these events are the mode of action leading to a decrease incidence of coronary heart disease.

Investigating a role for fibrates in the prevention of cardiovascular disease, a pooled meta-analysis of long-term randomized placebo-controlled clinical trials pointed out that fibrates significantly reduce the odds of nonfatal myocardial infarction, but not of cardiovascular mortality, fatal myocardial infarction, or stroke, compared with placebo (358). Fibrates can reduce the risk of major cardiovascular events predominantly by prevention of coronary events, and they might have a role in individuals at high risk of cardiovascular events and in those with combined dyslipidemia (359, 360). Furthermore, in a small-scale pilot study, bezafibrate was found to induce plaque regression in thoracic and abdominal aortas with marked TG reduction and HDL-C increase (361).

Human clinical trials and clinical trial endpoints

Primary endpoint results from the HHS and VA-HIT studies as well as the *post hoc* and analysis of the ACCORD and FIELD trials suggest that fibrates may benefit patients with high TG (≥200 mg/dL) and low HDL-C (329, 332). This hypothesis is now formally being tested in a phase 3 study with pemafibrate, called "Pemafibrate to Reduce Cardiovascular Outcomes by Reducing Triglycerides in Patients With Diabetes" (PROMINENT).

Clinical trials have found that fibrate treatment of atherogenic dyslipidemia reduces vascular atherosclerosis and cardiovascular risk (362). In the Bezafibrate Coronary Atherosclerosis Intervention Trial, bezafibrate treatment decreased angiographic evidence of coronary atherosclerosis (363, 364). Moreover, in the HHS, gemfibrozil decreased cardiovascular events, especially among patients with diabetes (354). Data from the VA-HIT (365), as well assubgroup analyses from the Bezafibrate Infarction Prevention Trial (366), have demonstrated that fibrate therapy significantly reduces the risk of cardiovascular events in particular in patients with diabetes with raised TGs and low HDL-C. However, large, well-powered interventional studies resulted in disappointing outcomes. The FIELD trial saw no benefit of fenofibrate, with the intervention group showing only nonsignificant differences compared with the controls (329). One potential explanation for this failure was the relatively high drop-in usage of statins in the placebo group and the lower overall rates of cardiovascular events. The ACCORD study looked at addition of fenofibrate to simvastatin, but it enrolled patients with little dyslipidemia (367–370). The currently running phase 3 trial PROMINENT will test the efficacy of pemafibrate in diabetic patients with high baseline TG levels (371). Meanwhile, fibrates are mainly prescribed for the management of raised TGs in patients at risk for pancreatitis.

One clinical feature noted through many of the fenofibrate studies was reduction in markers of systemic inflammation, including C-reactive protein (CRP) and IL-6. These reductions occurred even in populations lacking a significant change in lipid parameters, suggesting that the effect was direct and not mediated through changes in lipid-mediated vascular damage (372). This phenomenon may be the physiological consequence of immune cell PPAR activation and cross-talk/inhibition of NF-κB signaling. The combination of anti-inflammatory efficacy along with promotion of hepatic FAO suggested that PPARα activation may prevent the progression of liver steatosis to NASH and ultimately cirrhosis. The dual

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PPARα/δ agonist elafibranor was tested for efficacy in patients with liver steatosis. A reduction in histological endpoints of acute NASH was seen, although the trial did not achieve its originally set target of NASH resolution (195). Therapeutic efficacy of elafibranor is currently assessed in the phase 3 trial RESOLVE-IT, enrolling patients with active NASH and fibrosis stage F1 to F3 (373).

SPPARMαs in humans

Overall it is surprising that there are so few therapeutic applications for fibrates, considering the important role that PPARα has in regulating hepatic energy metabolism in response to the fasted state. Furthermore, although we have good evidence for cross-talk between PPARα and inflammatory signaling pathways, including NF-κB, the fibrates have not found a role in the management of chronic inflammatory diseases.

In an attempt to maximize the therapeutic potential of PPARα as a drug target, SPPARMαs are being developed to gain greater pharmacological effect (374, 375). A restricted activation potential can be explained at the molecular level by a general decrease in affinity for cofactors due to partial activation, binding to different target gene promoters, and, but not limited by, the induction of a differential receptor conformation, which leads to a changed PTM status and the recruitment of a specific subset of PPAR cofactors, also according to their expression level in different organs (26, 374, 376).

Pemafibrate is an example of a high-affinity PPARα agonist with SPPARM properties that is currently in phase 3 clinical development (191). Pemafibrate was designed to possess higher PPARα activity and selectivity over existing PPARα agonists. Pemafibrate monotherapy induced atheroprotective effects in several murine models of atherosclerosis. As a multilevel strategy compound, pemafibrate acts on inflammation, lipid metabolism, and RCT (190, 377). Promising results were achieved with pemafibrate for the treatment of a severe complication of type 2 diabetes, that is, diabetic nephropathy, by targeting the pathways leading to renal oxidative stress (378). The first-line treatment of hypercholesterolemia, a common feature of atherosclerotic cardiovascular disease, is statins, which lower the levels of LDL-C. However, remaining residual risks, such as elevated TG levels and decreased HDL-C, underscore the necessity for complementary therapies. Pemafibrate was found superior to fenofibrate with regard to serum TG-lowering effect and hepatic and renal safety, when used during 24 weeks in adults with dyslipidemia (191). Consequently, in recent randomized, double-blind, placebo-controlled clinical trials in patients with dyslipidemia, statins were combined with the SPPARMα pemafibrate, which showed a favorable benefit-to-risk ratio during phase 2 clinical trials for the treatment of combined dyslipidemia (379, 380). The compound was used as an add-on therapy in combination with statin treatment to treat the residual cardiovascular risk. Next to decreasing TG levels, this drug combination targets also several other atherogenic lipoproteins, thereby improving apolipoprotein ratios and lowering LDL-C levels (379, 380). In patients with type 2 diabetes with hypertriglyceridemia, pemafibrate was also shown to ameliorate lipid abnormalities and was well tolerated (381).

Next to SPPARMs, combining selective RXR agonists with statins can be an alternative option for the treatment of atherosclerosis. Suboptimal doses of both treatments not only minimized dose-related adverse effects, but also resulted in synergistic beneficial effects on endothelial function, compared with each drug alone. The anti-inflammatory effects were mediated in part by increased RXRα–PPARα and RXRα–PPARγ interactions (382).

Finally, dual or pan-PPAR modulators with partial or selective activation profiles may be good alternatives, depending on the clinical application.

Metabolic disorders such as diabetes, atherosclerosis, hyperlipidemia, and obesity rarely occur in isolation, but usually arise in the same individuals. This complex phenotype is called the metabolic syndrome or syndrome X (4, 374, 383). Treatment of this aggregation of metabolic abnormalities, usually accompanied by an inflammatory context, would benefit from a more holistic approach, such as the application of a pan-PPAR agonist, to target the multiple facets of this complex disease phenotype.

"It may well be that PPARα exerts a context-dependent immune system priming role."

Inflammation-Priming and Inflammation-Resolving Molecular Actions of PPARα

Evidence suggests that PPARα can counter inflammation via multiple, distinct mechanisms (23, 384, 385). Both *in vitro* and *in vivo* studies identified PPARα influences on both acute and chronic inflammatory processes (156, 164, 355, 386).

The first indication of a beneficial role for PPARα in suppressing inflammation came from studies on LTB_4 action in PPARα-deficient mice. Compared with wild-type mice, the duration of LTB_4-induced ear-swelling responses was prolonged. This anti-inflammatory effect correlated with LTB_4 activation of PPARα, thereby identifying a feedback loop through PPARα to limit LTB_4 action. These findings implied that some of the anti-inflammatory actions of PPARα occur through upregulation of enzymes involved in ω- and β-oxidation pathways and thereby accelerated degradation of signaling lipids including LTB_4 (164). Subsequently, Poynter and Daynes (130) showed that PPARα-deficient murine splenocytes had NF-κB activity, and thereby produced more IL-6 and IL-12 in response to lipopolysaccharide (LPS), supporting PPARα-dependent anti-inflammatory effects

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in nonmetabolic tissues. Most recent research has focused on the anti-inflammatory actions of PPARα. Hence, most of the sections below discuss the many and diverse molecular mechanisms accounting for the anti-inflammatory effect of PPARα in various models (Figs. 4 and 5). However, under certain conditions PPARα induces target genes that are more consistent with a proinflammatory profile (see further below), suggestive of inflammation priming (Fig. 6) or sensitization, the phenomenon of sequential signals leading to cellular adaptation and affecting subsequent responses, specifically in a context of inflammation. The augmented second response may well be gene specific and transient, as has been described for the response of synoviocytes to a repeated treatment with proinflammatory cytokines (387). These consecutive signals may encompass proinflammatory mediators, as in the study above, but may also combine signals of a different nature. For example, the anti-inflammatory GC hormones transcriptionally upregulate several components of the innate immune system and thus prime proinflammatory responses (388–390), a circuit thought to contribute to more effective resolution of inflammation. We hypothesize that similar mechanisms may also apply to PPARα agonist action. Hence, an overall anti-inflammatory network may also comprise a component that is "pro-inflammatory" dependent on the specific inflammatory and cellular context. More work is needed to substantiate these speculations. Data that may support this concept, however limited, are discussed in the last subsection.

Effector cells mediating the anti-inflammatory activity of PPARα

The extent to which PPARα plays a decisive role in various immune cells in humans as compared with rodents remains a point of contention. Literature and public resources such as the human immune genome project indicate that PPARα expression is lower in human immune cells than in hepatocytes. This originally supported the concept that cell types such as hepatocytes with high PPARαs are likely contributing to the immune regulatory effects by providing secondary signals to the immune cells, but data are accumulating on functional PPARα effects in different immune cells. Additionally, it is not inconceivable that during an inflammatory response, a change in the PPARα expression profile of immune cells may turn a dormant factor into an active contributor to assist in blocking the inflammatory response. In support of that assumption, the well-known adrenal stress hormones GCs enhance PPARα transcript levels in T and B lymphocytes (129), at least in the mouse.

PPARα is not only expressed in undifferentiated human monocytes but also in differentiated human monocyte–derived macrophages and is transcriptionally active upon ligand binding. In TNF- and IFN-γ–activated macrophages PPARα can induce apoptosis (32). All three PPAR subtypes play important roles in human macrophages. For example, the anti-inflammatory properties of PPARγ have been connected to its ability as a driver of M2 polarization in human atherosclerosis (391).

Brocker *et al.* (324) recently compared hepatocyte-specific to macrophage-specific PPARα-null mice and found that the downregulation of the inflammatory cytokines IL-15 and IL-18 depended on macrophage PPARα, identifying Küpffer cell PPARα as an important mediator of the anti-inflammatory effects of PPARα agonists. Functional human PPARα was demonstrated following a global gene expression profiling experiment on human primary macrophages treated with specific PPARα agonists (323). The role of PPARs, FA signaling, and FAO within primary human immune cell types in a variety of inflammatory conditions requires further experimental study. Linking

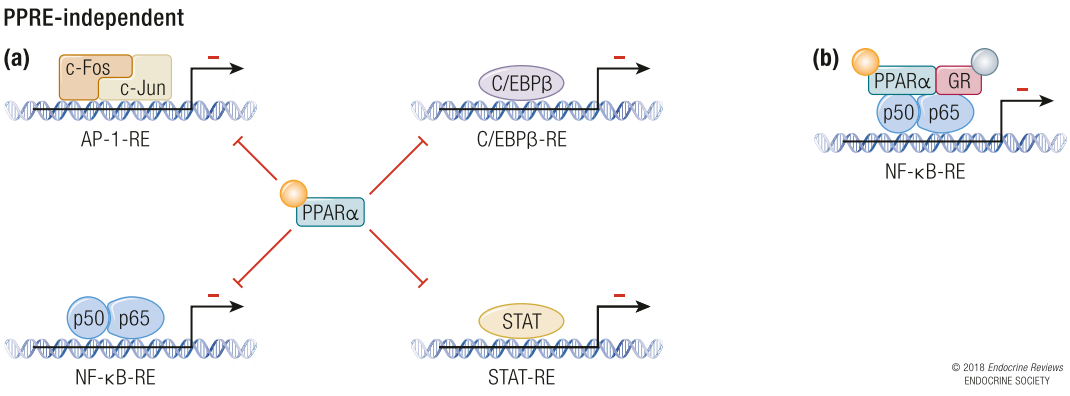


Figure 4. Models of PPARα transrepression activity. (a) PPARα directly interacts with proinflammatory transcription factors such as NF-κB, AP-1, and STAT to negatively regulate their targets, leading to anti-inflammatory effects through a DNA-independent mechanism. (b) Simultaneous ligand activation of PPARα and GR leads to the enhanced repression of TNF-induced IL-6 transcriptional activity, likely via a mechanism involving a direct physical interference between PPARα and GR. P, phosphorylated; PIASy, protein inhibitor of activated STATy; S, SUMOylated; Ub, ubiquitinated.

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inflammation to metabolism, murine macrophage LXR is critical for PPARα to promote cholesterol efflux and RCT *in vivo*, contributing to antiatherogenic effects and hence indirectly resolving inflammatory responses within the blood vessels. In support of this, PPARα activation was demonstrated to inhibit the development of atherosclerosis by promoting macrophage RCT in humanized hypercholesterolemic mice expressing the human APOA1 gene (392).

Connecting metabolic and immune responses to cancer, in murine and human melanoma models, an increase in free FA species during tumor progression was shown to activate PPARα signaling in $CD8^+$ tumor-infiltrating lymphocytes. Consequently, $CD8^+$ tumor-infiltrating lymphocytes switched toward FA catabolism to preserve their effector functions. Promoting FA catabolism with fenofibrate markedly improved the capacity of $CD8^+$ tumor-infiltrating lymphocytes to delay tumor growth via a synergistic mechanism involving a programmed cell death protein-1 blockade to efficiently boost the efficacy of melanoma immunotherapy (393). Taken together, these data highlight the interplay between the metabolic programs of T lymphocytes and tumor cells, with potential impact on melanoma immunotherapy (394). Ligand activation of PPARα also antagonizes NF-κB in murine T and B lymphocytes, and, conversely, T cell activation results in reduced PPARα expression. Of note, treatment of T lymphocytes with specific PPARα activators can upregulate the expression of endogenous PPARα-controlled genes in combination with histone deacetylase inhibitors. The presence of functional PPARα within T and B lymphocytes again emphasizes a role for this NR in cells of the immune system (129). In addition to targeting NF-κB, PPARα also counters proinflammatory signaling in T lymphocytes at the level of IL-2 and c-Jun activation (395). In evidence, genetic ablation of the PPARα gene resulted in the loss of antagonism of NF-κB, increased production of Th1, and decreased production of Th2 cytokines by T cells (395). In another study, the PPARα agonist Wy-14,643 was shown to increase IL-4 secretion and to inhibit IFN-γ expression in murine mitogen-activated splenocytes, and to reduce the proliferation of human T cell lines (396). Similarly, stimulation of isolated $CD4^+$ human T cells with PPARα agonists at clinically relevant concentrations decreased IFN-γ, TNF-α, and IL-2 production. However, in this study no effect on viability was noted (397).

Epidermal Langerhans cells (LCs) play a pivotal role in initiating and maintaining primary immune responses in the skin. PPARα is expressed in immature LCs but downregulated in mature LCs. Pharmacological PPARα activation inhibits LC maturation, migratory capacity, cytokine expression, and the ability to drive T cell proliferation. Moreover, PPARα in LCs also inhibits NF-κB activity. Taken together, PPARα activation by endogenous ligands may provide a molecular signal that allows LCs to remain in an immature state (398). The activation of PPARs by drugs, as well as products of intermediary energy metabolism within the diverse cells of the immune system, may reveal new opportunities to target inflammatory process or to advance adjunctive therapies to enhance anticancer treatment.

Interactions Between PPARα and other transcription factors

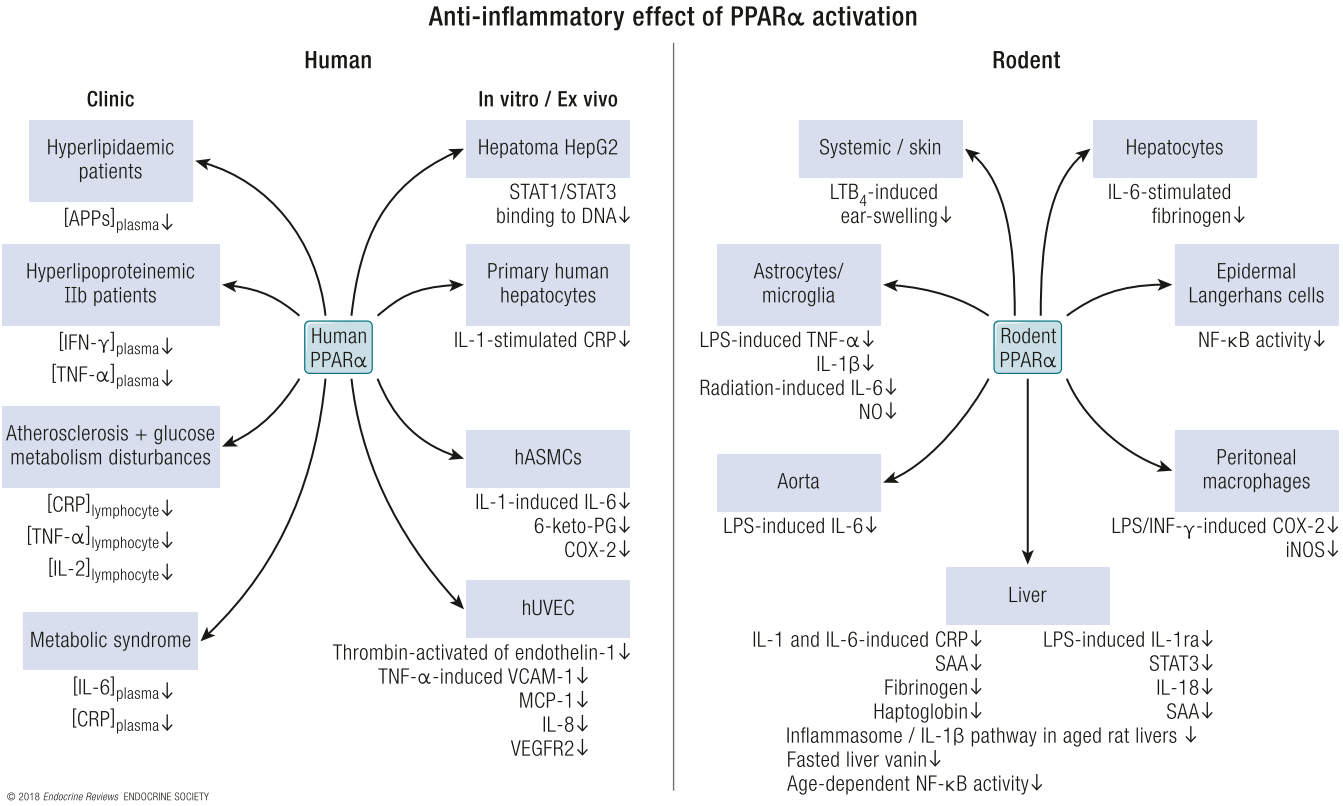
The best known mechanism by which PPARα (as well as the other PPAR subtypes) inhibits many inflammatory genes is transrepression. In this tethering mechanism, PPARα acts directly to negatively regulate the expression of proinflammatory genes by antagonizing the activities of other regulators of inflammation such as members of the NF-κB, AP-1, and STAT families, as also already alluded to in the section above (27) [Fig. 4(a)].

In this regard, it was found that fibrate-activated PPARα inhibits IL-1–stimulated secretion of IL-6 and 6-keto-prostaglandin by decreasing IL-6 and cyclooxygenase-2 (COX-2) gene transcription in smooth-muscle cells. Inhibition of COX-2 induction occurred transcriptionally as a result of PPARα interference with NF-κB signaling (23). This effect of PPARα in the vascular wall may influence the progress of atherosclerosis. Accordingly, Delerive *et al.* (355) have demonstrated that aortas from PPARα-null mice display an exacerbated inflammatory response to LPS, as assessed by enhanced IL-6 secretion, and that fibrate treatment represses IL-6 mRNA levels in LPS-stimulated aortas of PPARα wild-type mice, but not of PPARα-null mice, confirming that the observed anti-inflammatory properties of fibrates are PPARα-dependent. This vascular anti-inflammatory response is a result of PPARα interference with NF-κB and AP-1 transactivation capacity involving direct protein–protein interaction with p65 and c-Jun. Of note, a bidirectional antagonism between the PPARα and NF-κB signaling pathway exists (355).

In human vascular endothelial cells, PPARα inhibits the thrombin-mediated activation of endothelin-1, a vasoactive peptide. This inhibition occurs at the transcriptional level and occurs via interference with the AP-1 signaling pathway by preventing the binding of AP-1 proteins to their target sequence (399). Consistently, PPARα activators prevent TNFα–induced vascular cell adhesion molecule 1 expression in human endothelial cells, partly through inhibition of the NF-κB pathway. This PPARα action, with a consequent decrease in monocyte adherence to endothelial cells, may have important implications on the early stage of atherosclerosis (400). PPARα agonists decrease cytokine-induced tissue factor in monocytes most likely by interfering negatively with AP-1 and/or NF-κB activity (401, 402). Because these pathways are involved in the activation of inflammatory response genes

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Figure 5. Anti-inflammatory effects of PPARα activation. (left) Effect of human PPARα activation by various synthetical ligands on inflammatory mediators. (right) Effect of rodent PPARα on inflammatory mediators. Note that for some of these effects, conclusions on an involvement of PPARα were reached by using PPARα $^{-/-}$ mice, in which an aggravated inflammatory effect was observed. APP, acute phase response protein; hASMC, human aortic smooth muscle cell; hUVEC, human vascular endothelial cells; iNOS, inducible nitric oxide synthase; PG, prostaglandin; SAA, serum amyloid A; VCAM, vascular cell adhesion molecule.



implicated in atherogenesis, PPARα activation may be beneficial in combatting the pathogenesis of atherosclerosis. Additionally, oxidized eicosapentaenoic acid attenuates monocyte chemotactic protein 1 (MCP-1) and IL-8 expression in endothelial cells by inhibiting NF-κB in a PPARα-dependent manner (403).

The transcription factor specificity protein 1 (Sp1) is also involved in the anti-inflammatory mechanism of PPARα. PPARα activation inhibits vascular endothelial growth factor receptor-2 (VEGFR-2) in endothelial cells, which occur at the transcriptional level via decreasing Sp1 site-dependent binding to the promoter sequence through an increased interaction with the Sp1 transcription factor (404). Thus, PPARα activators may control inflammatory responses in part by targeting endothelial VEGFR-2 signaling, which is critical for angiogenic responses during chronic inflammation.

Further work has demonstrated that the mechanism of PPAR interference with NF-κB action additionally involves a PPARα-mediated induction of the inhibitory protein IκBα expression. Interestingly, fibrates increase IκBα expression in primary smooth muscle cells and hepatocytes, thus preventing nuclear p50/p65 NF-κB translocation and thereby arresting p65-mediated gene activation to oppose development of the inflammatory response. PPARα-mediated IκBα upregulation did not require a functional PPRE in the IκBα promoter, but rather involved functional Sp1 and NF-κB sites, in a cross-talk mechanism (405). Cytosolic retention of p50/p65 NF-κB extends the list of candidate anti-inflammatory PPARα mechanisms from those mainly characterized by direct binding and inactivation of transcription factors, such as p65/NF-κB and c-Jun (23, 355). Furthermore, in detailed molecular studies fibrates were found to decrease basal and IL-6–stimulated expression of the human fibrinogen gene in human primary hepatocytes and hepatoma HepG2 cells by interfering with the proinflammatory transcription factor C/EBP-β via sequestration of the coactivator GR-interacting protein 1/transcriptional intermediary factor 2 (406) [Fig. 4(a)]. This anti-inflammatory action of PPARα is not restricted to fibrinogen but also applies to other acute phase genes containing a C/EBP response element in their promoters (406).

Additionally, fibrate-activated PPARα negatively regulates IL-1–stimulated CRP gene expression in isolated primary human hepatocytes by two mechanisms, the inhibition of p50–NF-κB translocation and

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the downregulation of basal p50–NF-κB and C/EBP-β expression in a PPARα-dependent fashion (407). Such observations reveal a direct contribution of PPARα in controlling the inflammatory gene response in the liver. Furthermore, chronic PPARα agonist treatment inhibited the expression of IL-6–induced acute phase response (APR) gene expression *in vivo*. This suppression occurs transcriptionally owing to the downregulation of the IL-6 receptor components gp80 and gp130 in the liver, thereby reducing the phosphorylation and activation of the downstream transcription factors STAT3 and c-Jun that transduce the IL-6 signal, as well as to the reduction of the basal expression of the transcription factors C/EBP-α, -β, and -δ (408). Related to STAT signaling, van der Meer *et al.* (72) proposed that activated PPARα may downregulate gene expression in HepG2 cells by causing the loss of STAT1 and STAT3 binding to DNA. Therefore, interfering with STAT-dependent transcriptional activity, which has been previously well documented for other PPAR subtypes, may also contribute to the anti-inflammatory action of PPARα activation (72) [Fig. 4(a)]. Inhibitory effects of PPARα agonists on IL-1– and IL-6–induced expression of APR proteins, including CRP, serum amyloid A, fibrinogen, and haptoglobin, were also observed in mice displaying liver-restricted expression of PPARα (409).

PPARα reduced the stimulation of the APR mediated by LPS and prevented the LPS-induced increase of proinflammatory mediators IL-1, IL-6, and TNF in plasma via a liver-specific transcriptional mechanism. This resulted in decreased expression of the adhesion molecules intercellular adhesion molecule 1 and vascular cell adhesion molecule 1 in the murine aorta, suggesting that PPARα can control the systemic and vascular inflammatory response via a liver-specific action (409). Importantly, this implies that the spectrum of PPARα-controlled genes may be vastly larger in a more physiological context than in isolated cells (410). Further evidence of this comes from analysis in human liver slices where there was suppression of inflammatory genes, but not in primary hepatocytes. The most highly repressed genes included chemokines, IFN-γ–induced genes, and other immune-related genes (410).

Fibrate activation of PPARα inhibits hepatic inflammatory responses, which unimpeded lead to liver injury and progression to fibrosis, by a mechanism relying on transrepressive activity, rather than binding to cognate PPREs and activation of lipid metabolism genes (386). These findings indicate that PPARα agonists with dissociated activity, selectively modulating PPARα transrepression, might have the potential to prevent the progression of NASH to liver fibrosis through a direct counteraction of inflammatory responses independent of effects on intrahepatic lipid accumulation. In the same line, a recent report has shown that PPARα gene and protein expression are decreased in NASH (411). The PPARα levels were restored by pharmacological inhibition or genetic suppression of miRNA-21, a microRNA upregulated in the liver of patients with NASH, decreasing thereby liver injury, inflammation, and fibrosis (411). These findings suggest that inhibition of the miRNA-21 pathway might be an interesting new therapeutic strategy to combine with some of the current PPARα pharmacological agonists in NASH (411).

Finally, a PPRE-dependent mechanism of PPARα–NF-κB interference in the regulation of complement C_3 transcription has been observed in a human hepatoma cell line, identifying transactivation as another mechanism for the anti-inflammatory effects of PPARα [Fig. 2(b)]. Ligand-dependent activation of PPARα abolishes TNF-α–mediated upregulation of complement C_3 through a physical interaction of PPRE-bound PPARα with p65 within the C_3 promoter (412).

Cooperation between PPARα and other anti-inflammatory NRs

Simultaneous activation of PPARα and GRα, a well-characterized NF-κB repressor, enhances transrepression of NF-κB–driven gene expression and additively represses proinflammatory cytokine production, including MCP-1 and IL-6 in various cell lines (384) [Fig. 4(b)]. This finding offers a rationale for the control of inflammation by GRα and PPARα by inflammatory gene repression and suggests new approaches for the treatment of inflammatory diseases where combinations of PPARα and GRα activation could offer a more effective anti-inflammatory intervention.

This phenomenon is not restricted to PPARα because a similar proof of concept was recently also shown for PPARγ agonists in skin and lung. Indeed, house dust mite–induced skin sensitization, which aggravates asthmatic responses in mice via a mixed Th2/Th17 response, was prevented by treatment combining GCs with rosiglitazone (413). Similarly, the role of PPARα in estrogen-mediated anti-inflammatory activity in lungs was shown by the demonstration of a weakened response to estrogen in PPARα-deficient mice (414).

Regulation of MAPK and PKC kinase activities

Given that inflammatory responses are driven in part by the activation of MAPKs, PPARα ligands were also shown to modulate the degree of macrophage inflammation not only by reducing NF-κB activation, but also by reducing MAPK phosphorylation, hence effectively decreasing the expression of the proinflammatory mediator COX-2 and inducible nitric oxide synthase (162, 415).

PPARα was further shown to polarize T cell differentiation processes, also via a modulation of MAPK activities. In $CD4^+$ T lymphocytes, unliganded PPARα suppressed p38 MAPK phosphorylation. Ligand activation reversed this inhibition, augmenting the expression of the transcription factor of T cells (T-bet), a lineage-determinant of Th1 inflammatory responses (416) that drives IFN-γ expression. PPARα thus reduced

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the level of activation-induced IFN-γ produced by $CD4^+$ T cells. Mechanistically, the control of PPARα over T-bet expression occurs via a DNA-binding independent mechanism, mediated through the ability of unliganded PPARα to repress the phosphorylation of p38-MAPK following T cell activation. These data suggested a novel and important function for PPARα within T cells. Both in transformed T and B lymphocytes, PPARα ligands effectively repress NF-κB activity (129). Additionally, the anti-inflammatory effects of PPARα in macrophages involve an inhibition of the PKC signaling pathway (52, 156). Little is known about the exact molecular mechanisms underlying the PPARα-mediated modulation of (potentially other) kinase activities and to what extent this may contribute to an overall anti-inflammatory action.

Anti-inflammatory activity of PPARα in a PPRE-dependent manner

In addition to downregulating the expression of proinflammatory genes, PPARα also suppresses the inflammatory response by direct upregulation of genes with anti-inflammatory properties, suggesting a possible cooperation between PPARα-dependent transactivation and transrepression to govern anti-inflammatory pathways. Besides the upregulation of IκBα, discussed above, as a mechanism contributing to its anti-inflammatory actions (407, 417), activated PPARα strongly upregulated the hepatic expression of IL-1 receptor antagonist (IL-1ra), an inhibitor of cytokine signaling, during LPS-induced inflammation, whereas numerous inflammatory genes including $STAT_3$, serum amyloid A, and IL-18 were found to be downregulated (418). Transactivation and chromatin immunoprecipitation studies identified IL-1ra as a direct positive target gene of PPARα with a functional PPRE present in the promoter. Moreover, upregulation of IL-1ra by PPARα was conserved in human HepG2 hepatoma cells and the human monocyte/macrophage THP-1 cell line (418).

In the same context, heme oxygenase-1, a PPARα target gene with a functional PPRE, is also directly upregulated by PPARα ligands and contributes to the anti-inflammatory effects in human vascular cells (419).

Interestingly, the gene coding for vanin-1, a protein with anti-inflammatory potential, is robustly activated in liver by PPARα agonists, such as Wy-14,643, fenofibrate, and clofibrate (293), and functional PPRE sites have been identified in the vanin-1 promoter (420). Vanin activity prevents steatosis development in response to fasting, whereas its deficiency affects multiple genes involved in inflammation/immunity (421).

Considering the above studies, it is clear that PPARα may affect markers of inflammation in a direct or indirect manner. Below, we continue the discussion on recent findings regarding the role of PPARs in some specific examples of inflammatory disease models.

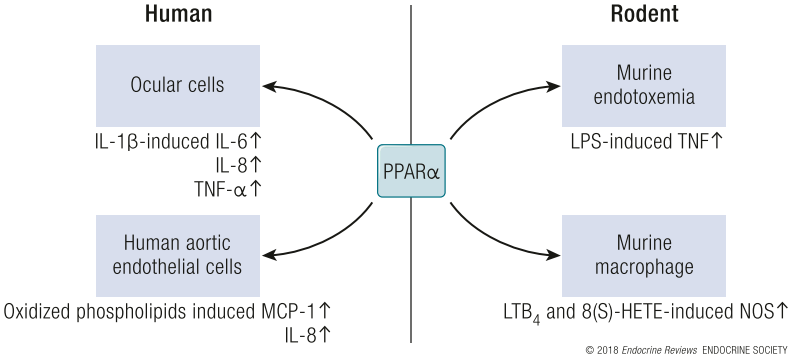


Figure 6. Stimulatory effects of human and murine PPARα activation on inflammatory mediators. Note here that mostly natural ligands seem able to trigger PPARα-dependent proinflammatory effects. NOS, nitric oxide synthase.

PPARα in the metabolic syndrome as an inflammatory disease

Inflammation is a critical component of the metabolic syndrome, which represents a cluster of risk factors that include dyslipidemia, insulin resistance, hypertension, inflammation, and coagulation disorders. Consequently, PPARα, being on the crossroad between inflammation and metabolism, is an important regulator of the metabolic syndrome (27, 422). It is now well accepted that proinflammatory cytokines and specific immune cell populations promote metabolic disease progression. Accordingly, a chronic low-grade inflammatory condition, which is characterized by increased circulating inflammatory cytokines and acute phase proteins and that often accompanies the metabolic syndrome, has been implicated as a major factor in both the establishment of the metabolic syndrome and its associated pathophysiological consequences (291). PPARα and other PPAR isoforms have all demonstrated anti-inflammatory effects in these conditions (423). In hyperlipidemic patients and patients with metabolic syndrome, PPARα activation using fenofibrate decreases plasma levels of several acute phase and inflammatory proteins that are normally increased during inflammatory conditions, such as high-sensitivity CRP and IL-6 (23, 372, 408). Additionally, fenofibrate treatment also reduces plasma concentrations of the cytokines IFN-γ and TNF-α in hyperlipoproteinemia type IIb patients (424). Addition of fenofibrate to simvastatin-treated patients with atherosclerosis and early glucose metabolism disturbances further reduces lymphocyte release of IL-2, IFN-γ, and TNF-α, accompanied by a reduction in plasma CRP levels, compared with statin alone (425). It was also found that fibrates have an anti-inflammatory effect in the interaction between adipocytes and macrophages in obese adipose tissue, thereby improving the dysregulated adipocytokine production and obesity-related metabolic syndrome (426).

PPARα in age-related inflammation

Enhancing PPARα levels and activity was shown to ameliorate age-related inflammatory processes.

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Increased inflammatory marker levels during aging correlate with a decreased PPARα activity. Administration of PPARα activators improved the splenocyte cellular redox balance and reduced the age-associated elevations in NF-κB activity and constitutive proinflammatory cytokine production (131, 427). The PPAR pan-agonist MHY2013 alleviated age-related hepatic lipid accumulation, in part by upregulating β-oxidation signaling and suppressing inflammation in the liver (428).

In a similar context, a recent study has demonstrated that LPS-induced liver inflammation and lipid accumulation were associated with a chronically overactive inflammasome–IL-1β pathway in aged rat livers and that these effects were due to the dysregulation of PPARα and SREBP1c. Indeed, in these aged rat livers, PPARα was consistently inactivated and the levels of SREBP1c were chronically increased (131). Therefore, targeting PPARα signaling might reduce the age-related sensitivity to endotoxin-induced liver inflammation.

These data, positioning PPARα at the crossroad of aerobic metabolism and inflammation, fit the observations that link nonaerobic glycolytic ATP generation to the production of proinflammatory cytokines (80). To illustrate, complex host responses to the TLR9 agonist CpGA were found to be mediated by plasmacytoid dendritic cell production of IFNs, operating in an autocrine fashion. FAO was necessary for full plasmacytoid dendritic cell activation, pointing to PPARα as a potential target to modulate downstream effects of type I IFNs (429).

PPARα in inflammatory bowel disease

Considering the intestine as an organ contributing to the immune system, a link between PPARα signaling and the control of colitis was found upon studying the IL-10 gene–deficient mouse as a model for human inflammatory bowel disease. In this model anti-inflammatory IL-10 gene expression changes were related to PPARα activation although the mechanisms responsible remain undefined (430). Infectious disease control in intestine also involves PPARα activation. As such, clearance of a *Giardia muris* infection was found to coincide not only with IL-17 production, contributing to protection, but also with an early PPARα induction (431). Along similar lines, fenofibrate represses IL-17 and IFN-γ expression and improves colitis in IL-10–deficient animals, suggesting a possible therapeutic potential in Crohn's disease (432). Of note, PPARα protects the intestine from colitis-induced permeability (433).

PPARα in inflammatory skin disease

PPARα activation impaired epidermal LC maturation and migratory capacity and alters cytokine production through the inhibition of NF-κB phosphorylation (398), suggesting a role for PPARα in combatting inflammatory skin diseases (434). In this regard, PPARα-deficient mice exhibit impaired recruitment of neutrophils and monocytes/macrophages to the wound bed and present a transient delay in healing, which coincides with the inflammatory phase (435). These data support the concept that PPARα favors skin healing via a modulation of the inflammatory phase (435). Additionally, PPARα activation is beneficial in mouse models with a hyperproliferative epidermis (438) and in models of irritant and allergic dermatitis (437). The anti-inflammatory properties of PPARα agonists, coupled with their antiproliferative and prodifferentiating effects, suggest that they could be beneficial for the treatment of a variety of cutaneous diseases (437). In a model of atopic dermatitis, topical application of the PPARα agonist WY14643 significantly decreased antigen-induced skin inflammation, providing *in vivo* evidence for the role of PPARα in controlling skin inflammation and pointing to a potential benefit in combating systemic inflammatory responses in patients with atopic dermatitis, a disease characterized by inflammatory infiltrates and barrier dysfunction (438). Interestingly, PPARα may be the molecular mediator of the antiallergic and anti-inflammatory effects of palmitoylethanolamide, a natural FA derivative present in murine skin (439). Taken together, the many observations above underscore the implications of PPARα in inflammatory skin disorders and pave the way for trials of topical PPARα agonist application.

PPARα in other inflammatory diseases

Expanding the applicability of PPARα agonists to other inflammatory diseases, anti-inflammatory properties of activated PPARα were observed in primary murine astrocytes, a cell type implicated in the pathology of experimental autoimmune encephalomyelitis, an animal model of multiple sclerosis. Fibrates decreased the LPS-induced secretion of proinflammatory cytokines TNF-α, IL-1β, and IL-6 and nitric oxide production, likely via interference with the DNA binding of NF-κB (440), suggesting a putative function of PPARα in controlling neurodegenerative diseases. In the same line, PPARα activation significantly inhibited radiation-induced microglial production of proinflammatory cytokines such as TNFα, IL-1β, Cox-2, and inducible nitric oxide synthase by inhibiting nuclear translocation of the NF-κB/p65 subunit and reducing phosphorylation of nuclear c-Jun/AP-1 (441). Genetic ablation of PPARα in an experimental model of autoimmune encephalomyelitis increased clinical symptoms in male but not female SV.129 H^{2b} mice, consistent with the finding that male mice of this strain express higher levels of PPARα than do female mice (395). Recently, PPARα was reported to contribute to the anti-inflammatory activity of melatonin in spinal cord injury, again by modulating the NF-κB pathway. It was

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demonstrated that melatonin was less effective in PPARα-null mice, compared with wild-type mice, as evaluated by inhibition of the degree of spinal cord inflammation and tissue injury, neutrophil infiltration, proinflammatory cytokine expression, NF-κB activation, and inducible nitric oxide synthase expression (442).

The paradox of PPARα as a proinflammatory mediator

The above studies (Fig. 5) are in line with the well-accepted notion that PPARα is involved primarily in anti-inflammatory signaling. As stated in the introduction to this section, PPARα has also been ascribed potential "pro-inflammatory" effects (Fig. 6). Lee *et al.* (443) suggested that oxidized phospholipids induce MCP-1 and IL-8 expression through a PPARα-dependent signaling cascade in human aortic endothelial cells. These data demonstrate that PPARα can mediate the activation of endothelial cells by stimulating the production of monocyte chemotactic activity in response to oxidized phospholipids and lipoproteins (444).

Additionally, PPARα activation by Wy-14,643 enhances IL-1β–induced proinflammatory cytokine release, including IL-6, IL-8, and TNF-α, in ocular cells and induces proangiogenic responses (444). In contrast, in the FIELD and ACCORD studies (329, 332) fibrates showed protection against diabetic retinopathy. Concordantly, the antiangiogenic activity of PPARα ligands specifically depended on the activation of PPARα because PPARα ligands inhibited FGF2 corneal neovascularization in wild-type but not in PPARα-deficient mice (445). Consistent with these data, fenofibrate was shown to improve diabetic retinopathy in type 1 diabetes animal models and ischemia-induced retinal neovascularization. Herewith, PPARα dependency also formed the mechanistic basis for the therapeutic effects of fenofibrate (446). The anti-inflammatory effects of fenofibrate-activated PPARα were ascribed to the inhibition of NF-κB signaling in the diabetic retina (446). The basis for these differences remains unclear but might reflect a reduced complexity of *in vitro* studies vs *in vivo* studies.

Then again, in a mouse model of endotoxemia, fenofibrate administration strongly increased circulating LPS-induced TNF levels (447). In a rat hypertensive model, bezafibrate was shown to sensitize for endotoxin effects, concomitant with low levels of HDL and increased levels of TG (448). In murine macrophage lines and in contrast to Wy-14,643, the natural PPARα ligands LTB_4 and 8(*S*)-HETE showed a proinflammatory potential by increasing NOS activity (347). These observations are again not restricted to PPARα, because PPARγ activity in T cells and dendritic cells promotes Th2 immunity (449, 450). These are interesting findings suggesting that PPARα activation may result in different responses depending on the inflammatory settings, an effect that may involve combinatorial signaling. Although the mechanisms by which such potentially "pro-inflammatory" effects occur is not understood, these data point to the need for a careful evaluation of the effects of PPARα agonists in the relevant cell types, and in different physiological contexts. How to reconcile these seemingly contradictory findings with the wealth of data on anti-inflammatory PPAR actions is as of yet unclear. For instance, reduced PPARα expression has also been linked to a lower overall survival and an enhanced tissue bacterial load in sepsis (451). The timing and nature of the proinflammatory stimulus and/or treatment may play a decisive role. It may well be that PPARα exerts a context-dependent immune system priming role, as has been documented recently also for GCs (389). Given that GR and PPARα are able to physically interact and able to cocontrol genes involved in both inflammatory and metabolic pathways (159, 384) increases the likelihood of this assumption. More research is clearly necessary to disentangle and explain these paradoxical observations. This hitherto hidden role of PPARα not only merits further mechanistic attention but also requires a better understanding at the level of physiological relevance.

Future Perspectives

On account of their pleiotropic effects, PPARs may be targets in a number of disease conditions. It is clear there is still a lot of ground to cover, foremost by studying novel generations of more potent and better tolerated single, dual, or pan-PPAR agonists in the clinic. More basic research involving novel PPARα agonists is important, as many mechanistic aspects differ from the first generation of PPARα agonists. Plus, novel fields of application may still arise, for which the first generation of drugs did not show therapeutic benefit.

Fibrates might be beneficial in the prevention of coronary heart disease. The highest benefit is achieved in cardiovascular disease patients with high TGs and low HDL-C (452). A clinical benefit of PPARα activation with novel agonists remains to be demonstrated in patients with high TGs/low HDL and residual cardiovascular risk. Fibrates might be useful in the treatment and prevention of atherosclerosis, and especially the new fibrates with greater potency open new perspectives. However, it remains unclear whether fibrates in general are useful in the treatment or prevention of heart failure.

It is now evident that the effects of PPARα are much more widespread and cover numerous aspects of inflammation and inflammatory-related diseases;

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however, many mechanistic fundamental details of its function still remain to be unraveled. For instance, NR regulation of circadian rhythms in inflammatory metabolic diseases deserves more clinical investigation. There is newfound appreciation for the timing of treatment interventions. For example, a recent study shows a link between the circadian clock and perioperative myocardial injury in patients undergoing aortic valve replacement, whereby antagonism of Reverbα, another NR family member and PPARα target gene (271), seems to provide a benefit for cardioprotection (453). Moreover, the circadian clock regulates the expression and activity of genes and proteins involved in the progression of several autoimmune diseases (454, 455). Future therapeutics relying on PPARs in a context of inflammation will therefore most probably have to take "the clock" into consideration.

The role of PPARα needs to be further evaluated in various pathophysiological conditions. For instance, it is unclear how PPARα may influence immune cell (metabolic) control in immune/inflammatory diseases that are GC resistant, such as sepsis.

Defining the molecular mechanisms of PPARα function that govern inflammatory programs will improve our understanding of its roles and will allow us to further exploit their potential in the therapeutic field.

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Abbreviations

ACCORD, Action to Control Cardiovascular Risk in Diabetes; ACOX, acyl–coenzyme A oxidase; AF-1, activation function-1; AP-1, activator protein-1; APOA1, apolipoprotein A-I; APOA2, apolipoprotein A-II; APOA5, apolipoprotein A-V; APOC3, apolipoprotein C-III; APR, acute phase response; AMPK, AMP-activated protein kinase; CBP, cAMP response element binding protein; C/EBP, CCAAT/enhancer-binding protein; CoA, coenzyme A; COX-2, cyclooxygenase-2; CPT1, carnitine palmitoyl transferase 1; CREBH, cAMP-responsive element-binding protein H; CRP, C-reactive protein; Cyp7a1, cholesterol 7α-hydroxylase; DBD, DNA-binding domain; DR-1, direct repeat type 1; ERR, estrogen-related receptor; FA, fatty acid; FAO, fatty acid oxidation; FAS, fatty acid synthase; FGF21, fibroblast growth factor 21; FIELD, Fenofibrate Intervention and Event Lowering in Diabetes; FOXO1, forkhead box protein O1; GC, glucocorticoid GR, glucocorticoid receptor; GSK3, glycogen synthase kinase 3; HDL, high-density lipoprotein; HDL-C, high-density lipoprotein C; HETE, hydroxyeicosatetraenoic acid; HFD, high-fat diet; HNF4, hepatocyte nuclear factor 4; HHS, Helsinki Heart Study; hPPAR, human peroxisome proliferator–activated receptor; Hsp90, heat shock protein 90; IL-1ra, IL-1 receptor antagonist; LBD, ligand-binding domain; LBP, ligand-biding pocket; LC, Langerhans cell; LDL, low-density lipoproteinLDL-C, low-density lipoprotein cholesterol; LPL, lipoprotein lipase; LPS, lipopolysaccharide; LTB_4, leukotriene B_4; LXR, liver X receptor; MCD, malonyl–coenzyme A decarboxylase; MCP-1, monocyte chemoattractant protein 1; MHC, myosin heavy chain; mHMG-CoA, mitochondrial 3-hydroxy-3-methylglutaryl–coenzyme A; NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; NCoR, nuclear receptor corepressor; NF-κB, nuclear factor κB; NR, nuclear receptor; PBP, PPAR-binding protein; PGC, peroxisome proliferator–activated receptor γ coactivator; Pck1, phosphoenolpyruvate carboxykinase; PKA, protein kinase A; PKC, protein kinase C; PPAR, peroxisome proliferator–activated receptor; PPRE, peroxisome proliferator–activated receptor response element; PTM, posttranslational modification; RCT, reverse cholesterol transport; RXR, retinoid X receptorSIRT1, silent information regulator 1; Sp1, specificity protein 1; SPPARM, selective peroxisome proliferator–activated receptor modulator; SPPARMα, selective peroxisome proliferator–activated receptor α modulator; SRC-1, steroid receptor coactivator-1; SREBP, sterol regulatory element–binding protein; STAT, signal transducer and activator of transcription; SUMO, small ubiquitin-related modifier; TG, triglyceride; UTR, untranslated region; VA-HIT, Veterans Affairs HDL Intervention Trial; VEGFR-2, vascular endothelial growth factor receptor-2; VLDL, very-low-density lipoprotein.

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